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Goto

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(54) **COUPLER AND CALCULATING DEVICE**

(56) **References Cited**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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This patent is subject to a terminal disclaimer.

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Primary Examiner — William Hernandez

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Related U.S. Application Data

(62) Division of application No. 18/348,676, filed on Jul. 7, 2023, now Pat. No. 12,081,185, which is a division
(Continued)

(57) **ABSTRACT**

According to one embodiment, a coupler includes first to fourth capacitors, first and second inductors, and a first Josephson junction. The first capacitor includes a first capacitor end portion and a first capacitor other-end portion. The first inductor includes a first inductor end portion, and a first inductor other-end portion. The second inductor includes a second inductor end portion, and a second inductor other-end portion. The first Josephson junction includes a first Josephson junction end portion, and a first Josephson junction other-end portion. A space is surrounded with the first inductor, the second inductor, and the first Josephson junction. The third capacitor includes a third capacitor end portion, and a third capacitor other-end portion. The fourth capacitor includes a fourth capacitor end portion, and a fourth capacitor other-end portion.

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(51) **Int. Cl.**

H03H 11/02 (2006.01)

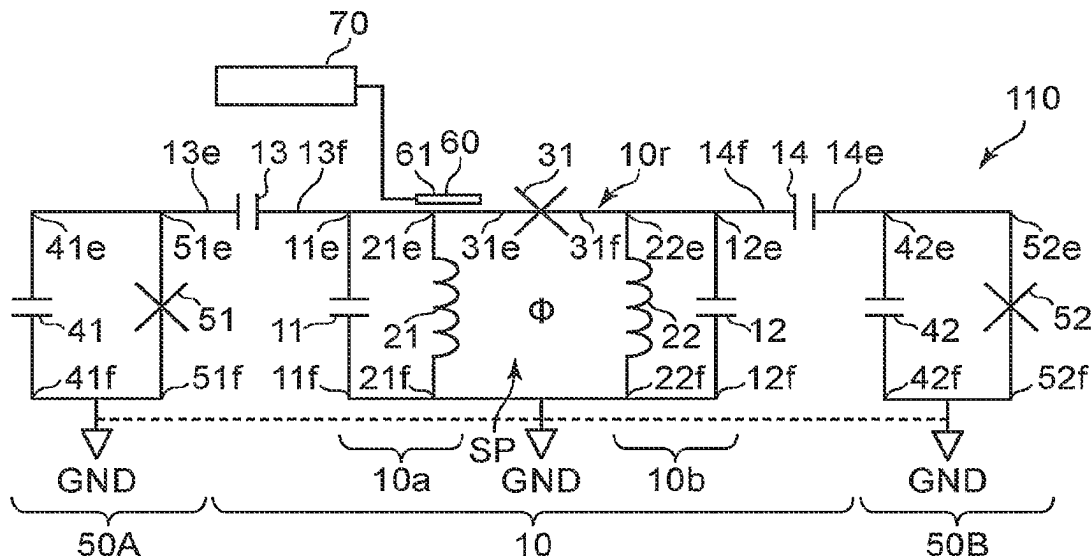
(52) **U.S. Cl.**

CPC **H03H 11/02** (2013.01); **Y10S 505/865** (2013.01)

(58) **Field of Classification Search**

CPC .. H03H 11/02; Y10S 505/864; Y10S 505/865
(Continued)

18 Claims, 22 Drawing Sheets



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of application No. 17/931,515, filed on Sep. 12, 2022, now Pat. No. 11,742,831, which is a division of application No. 17/681,610, filed on Feb. 25, 2022, now Pat. No. 11,476,836.

(58) **Field of Classification Search**

USPC 327/528, 594
See application file for complete search history.

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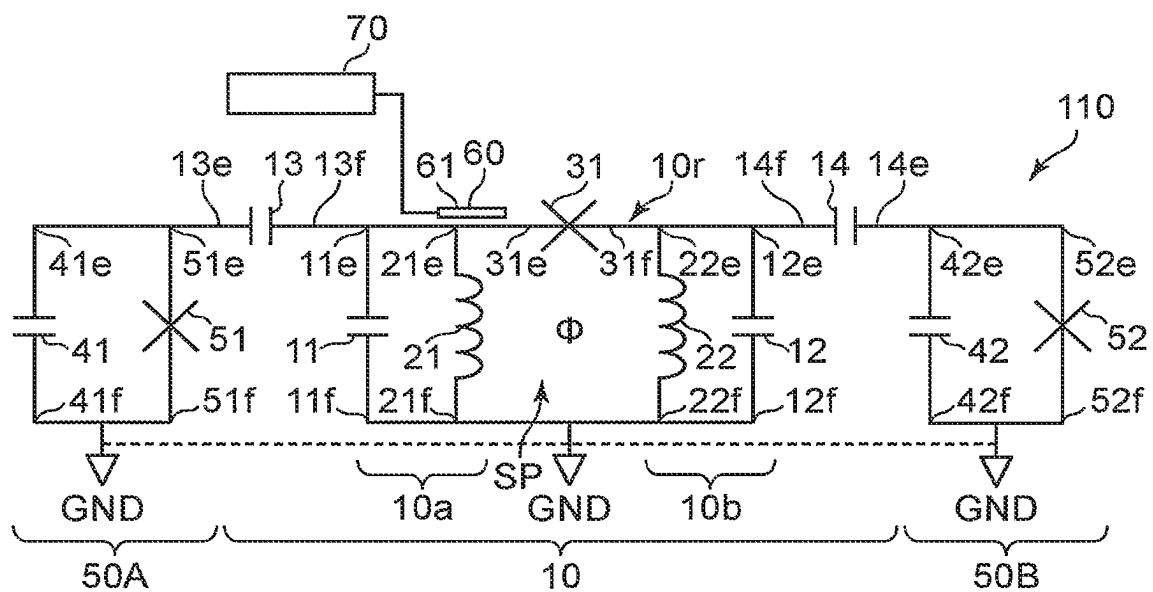


FIG. 1

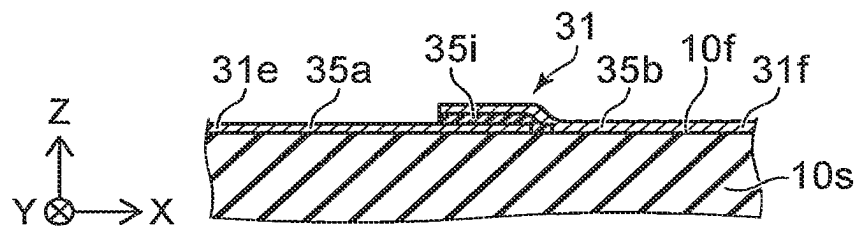


FIG. 2A

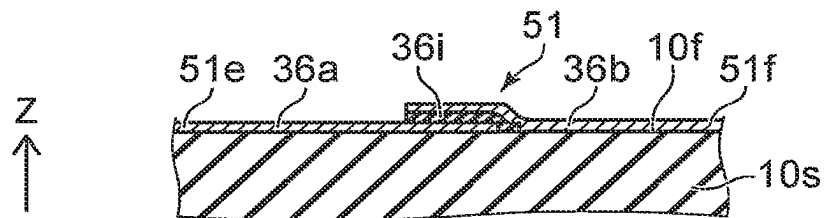


FIG. 2B

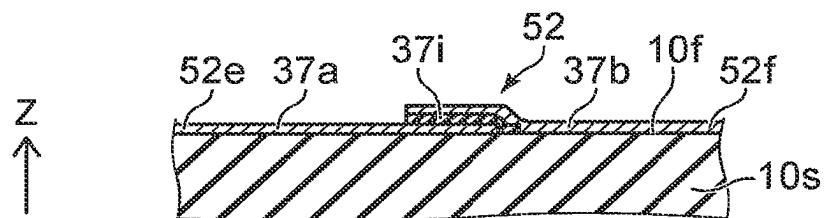


FIG. 2C

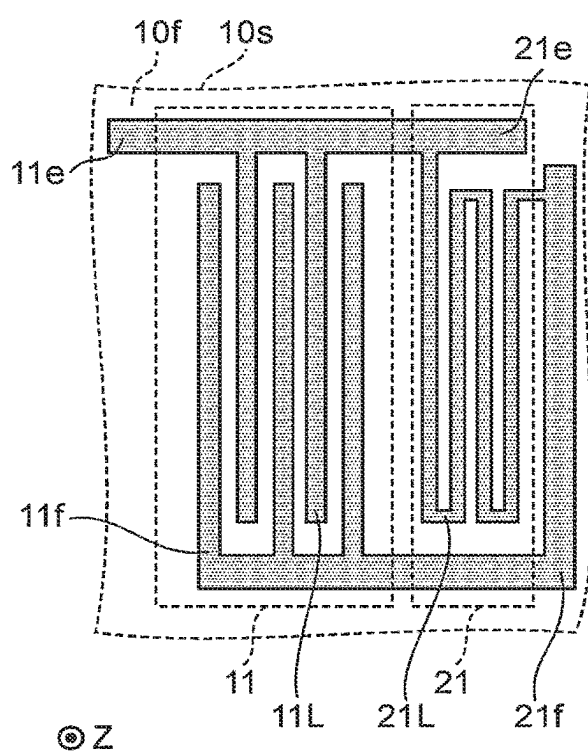


FIG. 3A

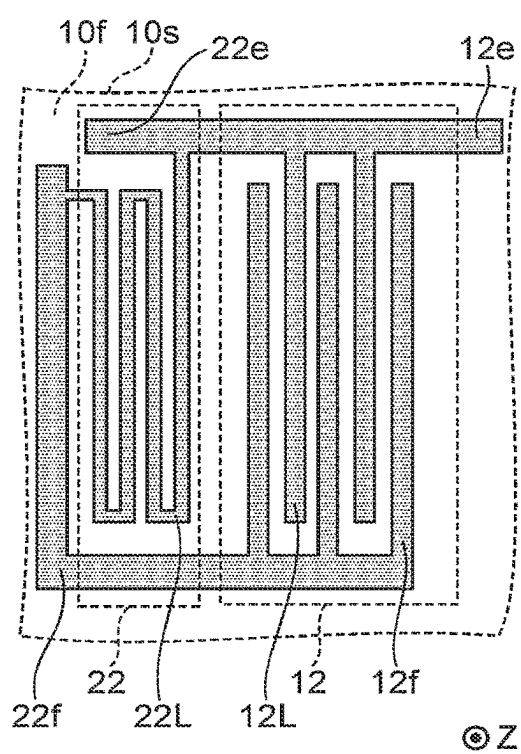


FIG. 3B

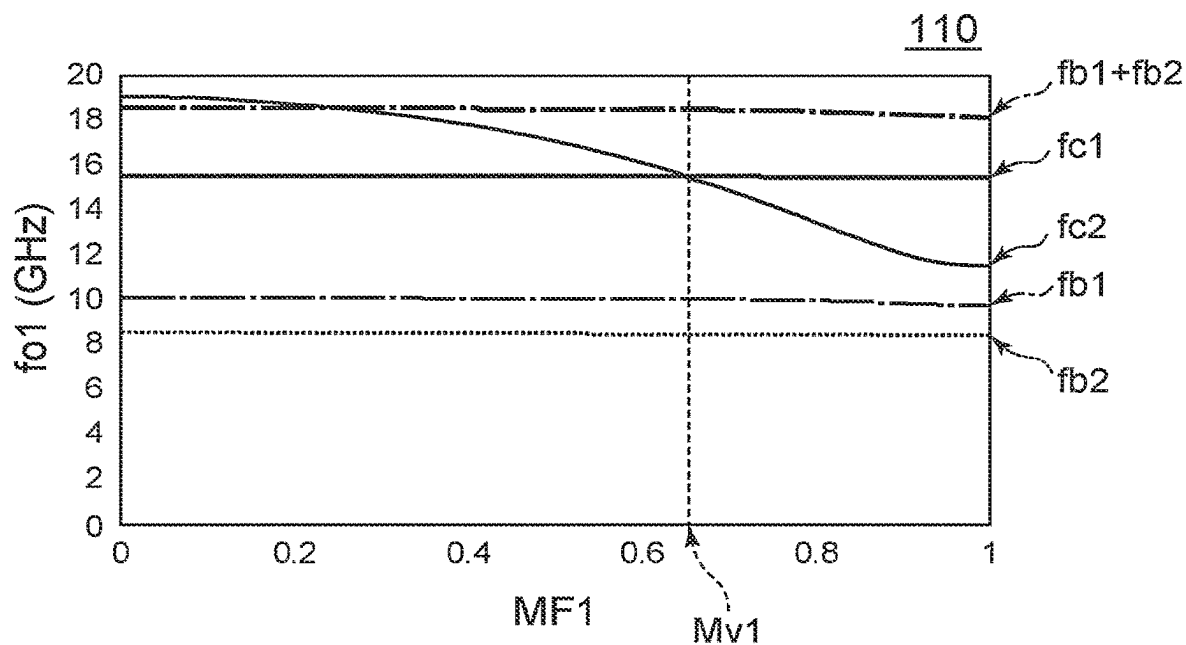


FIG. 4A

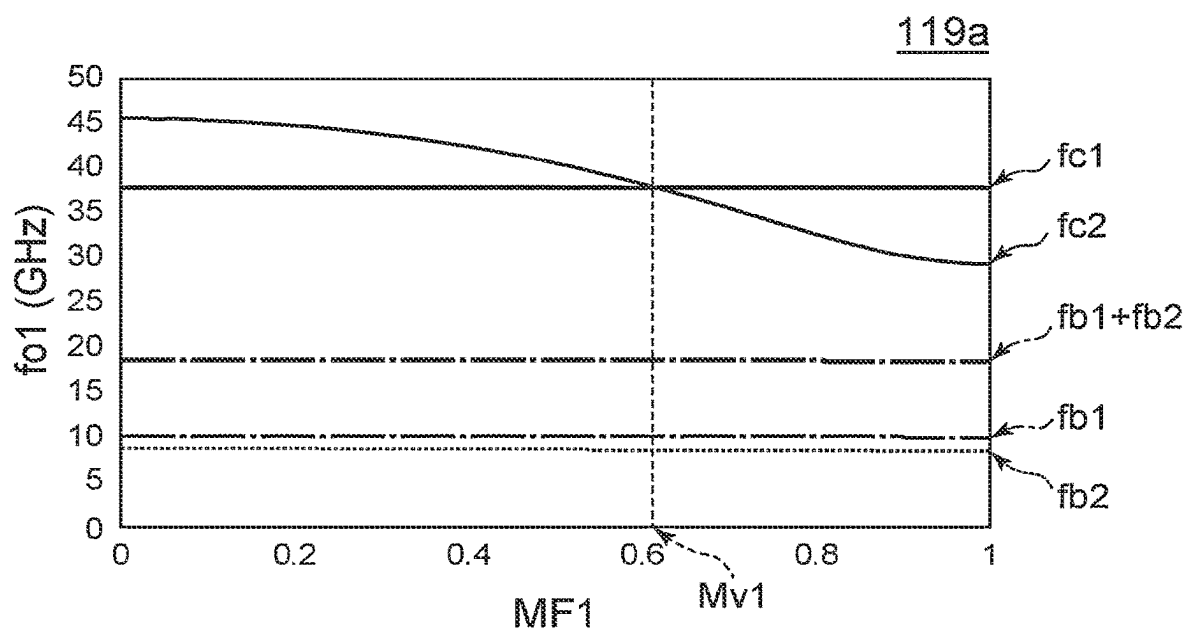


FIG. 4B

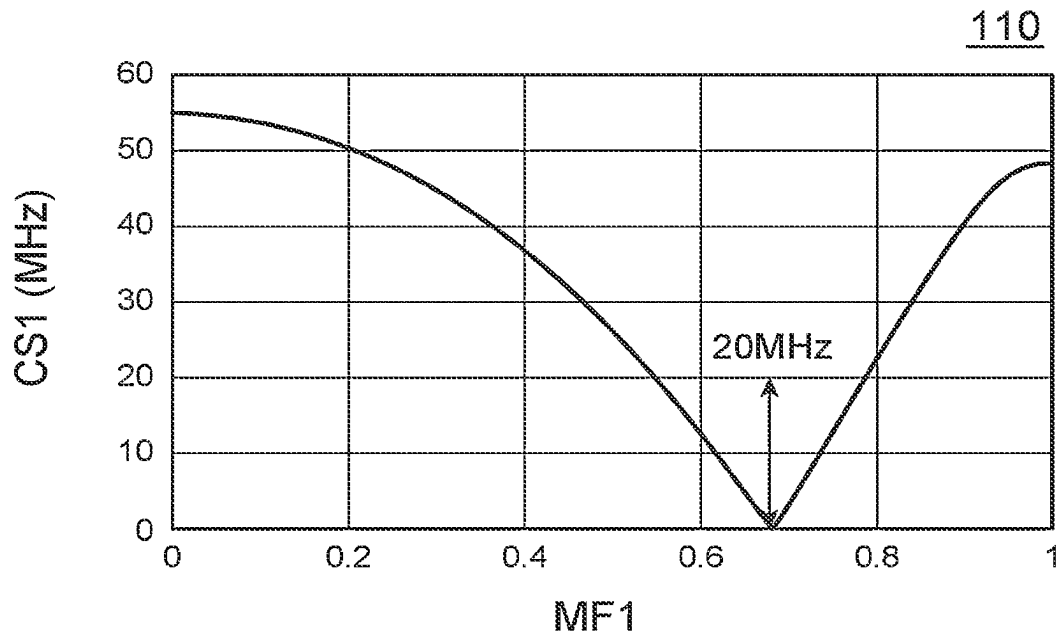


FIG. 5A

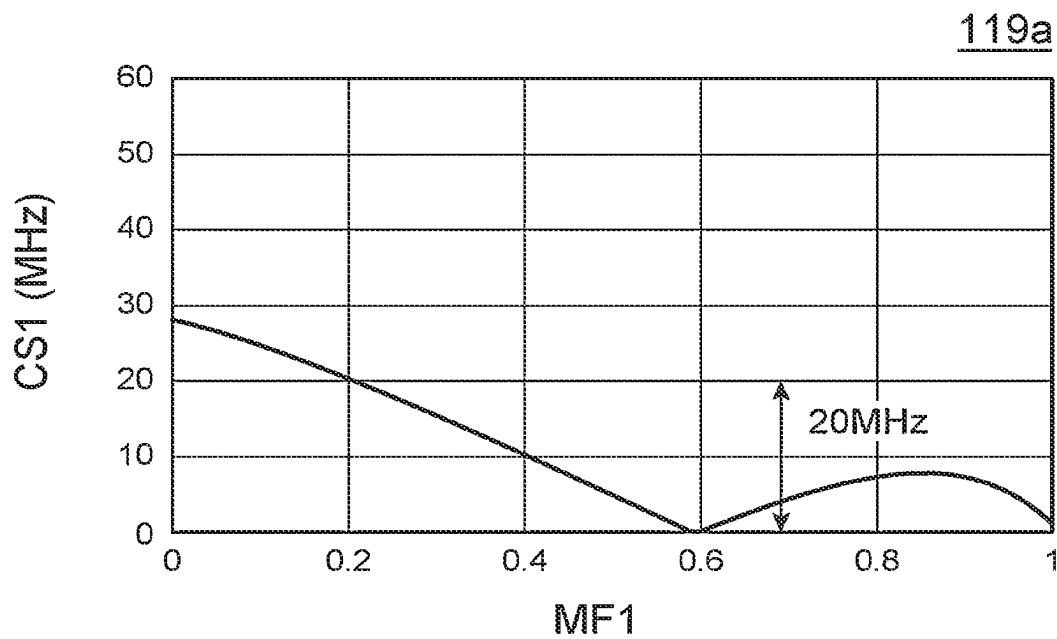


FIG. 5B

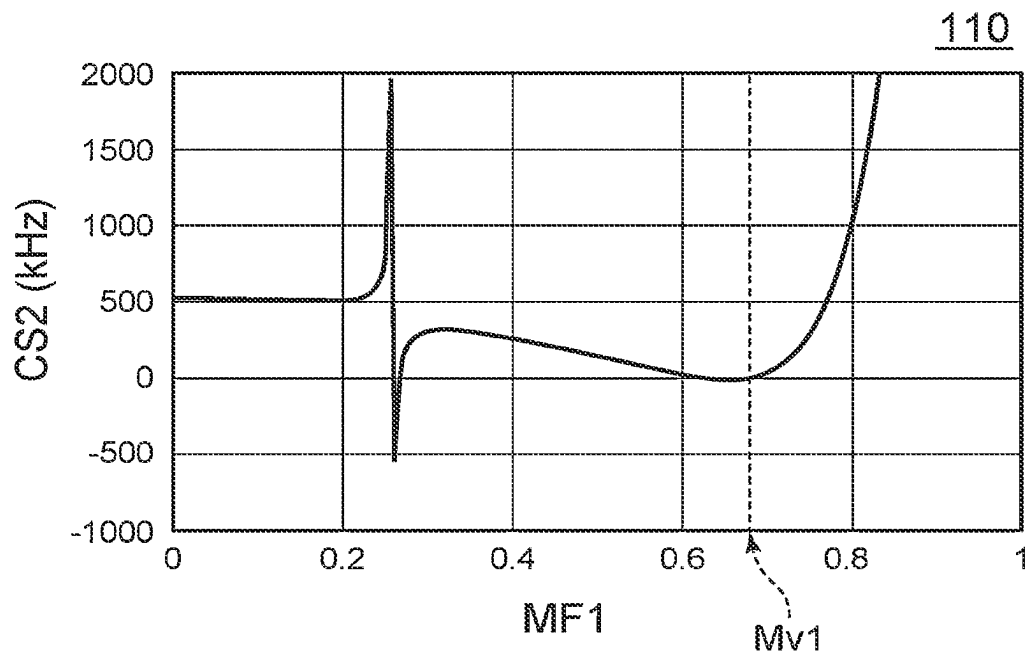


FIG. 6A

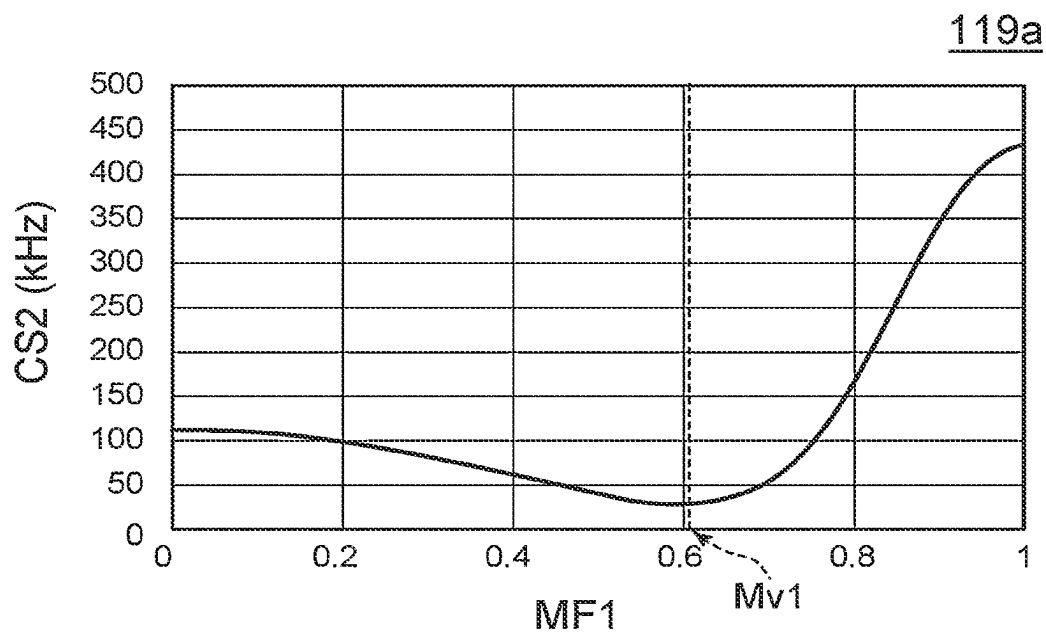


FIG. 6B

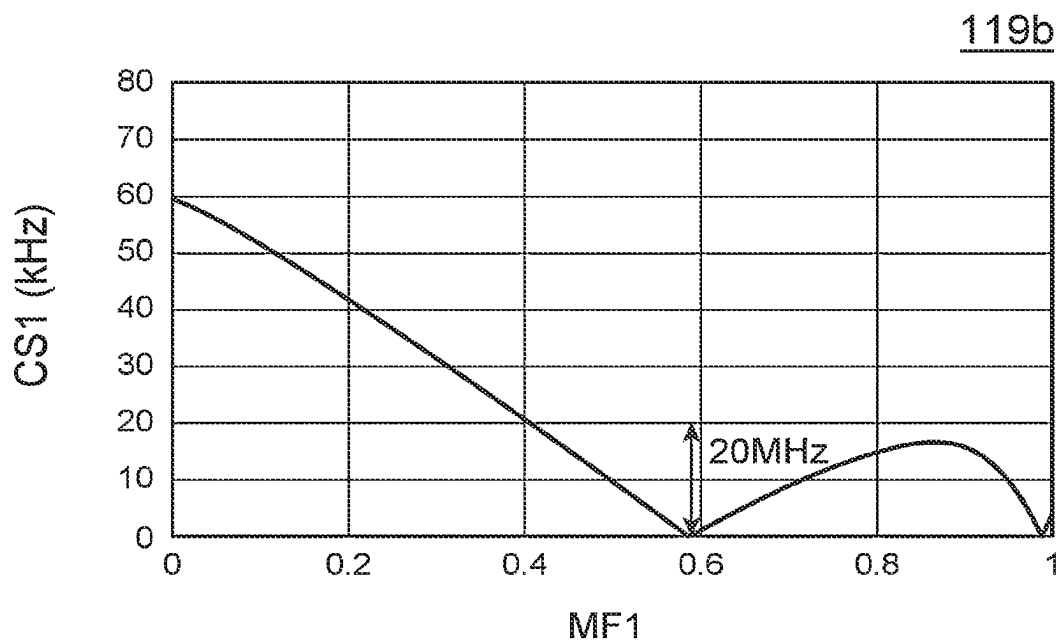


FIG. 7A

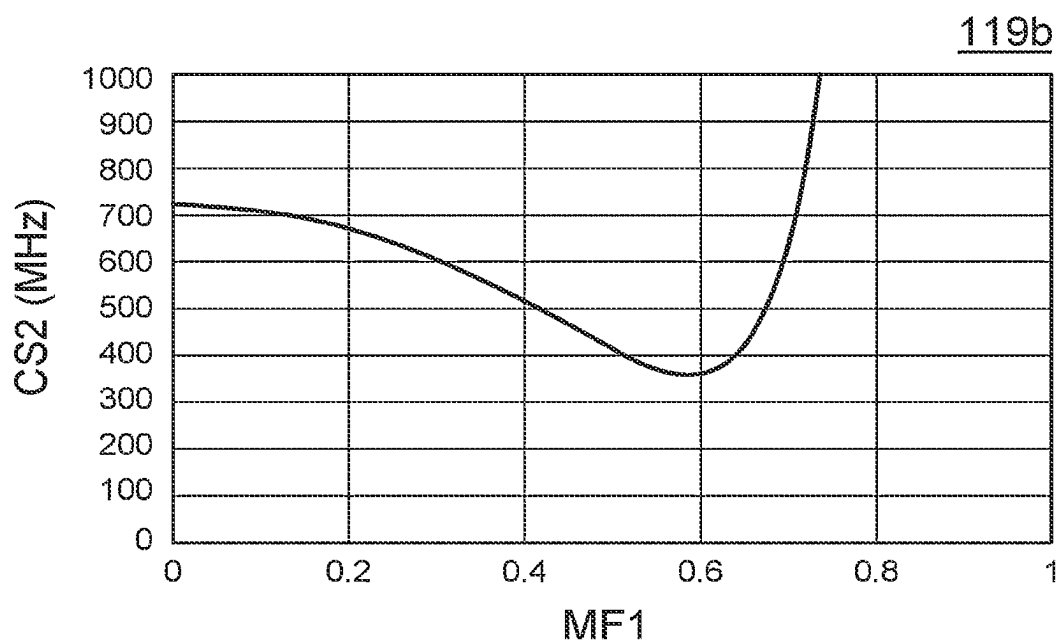


FIG. 7B

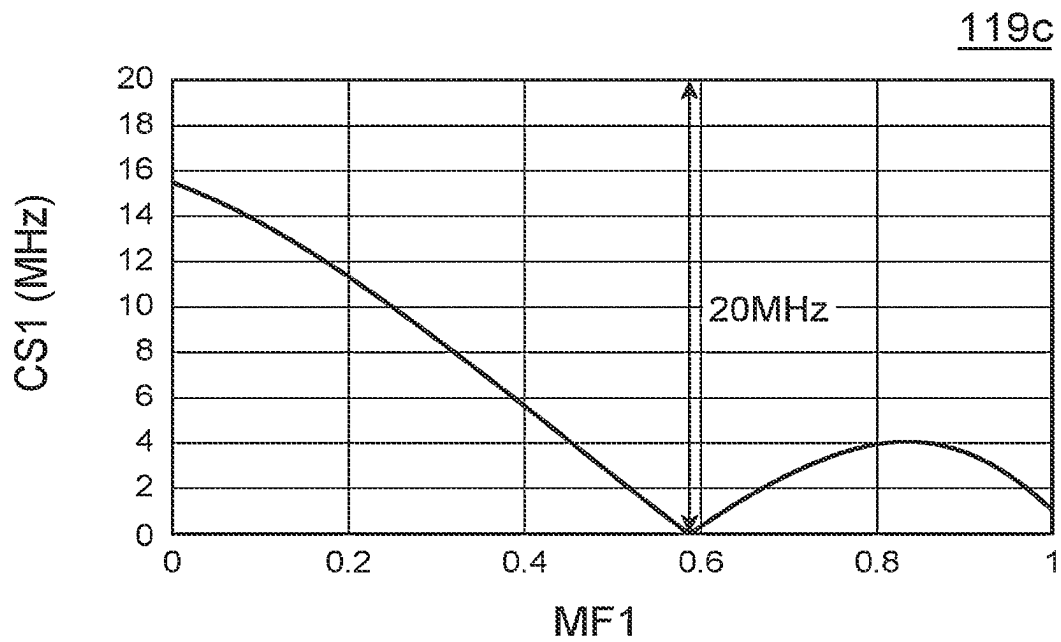


FIG. 8A

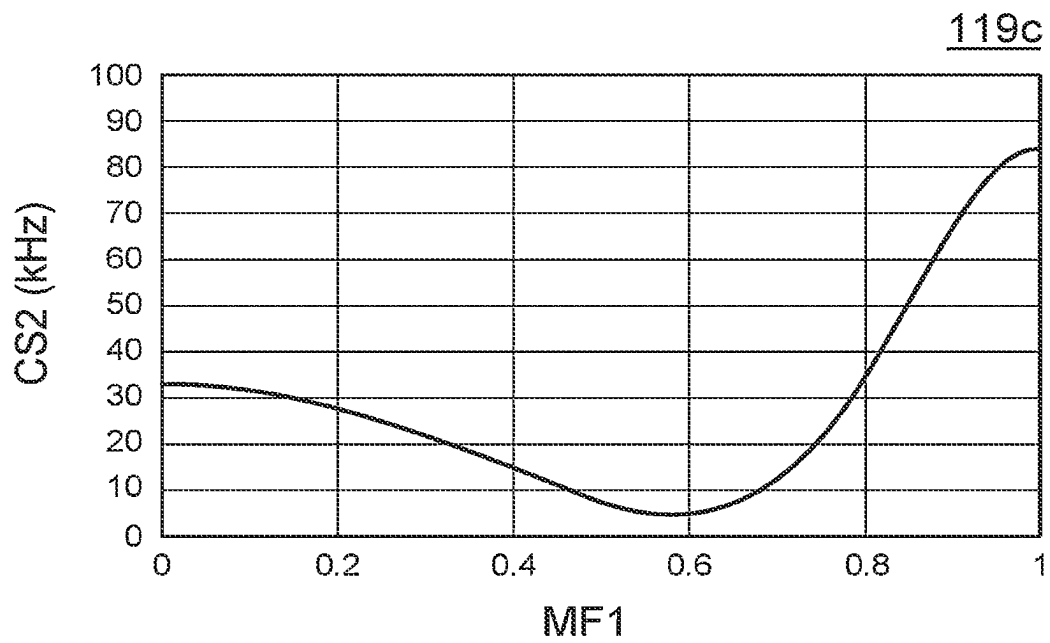


FIG. 8B

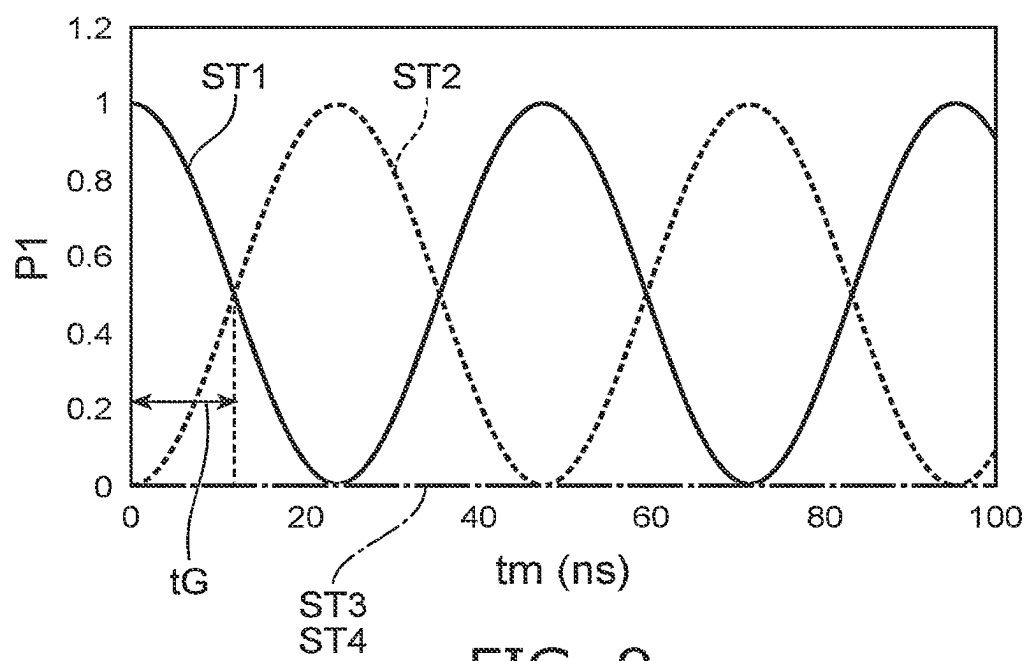


FIG. 9

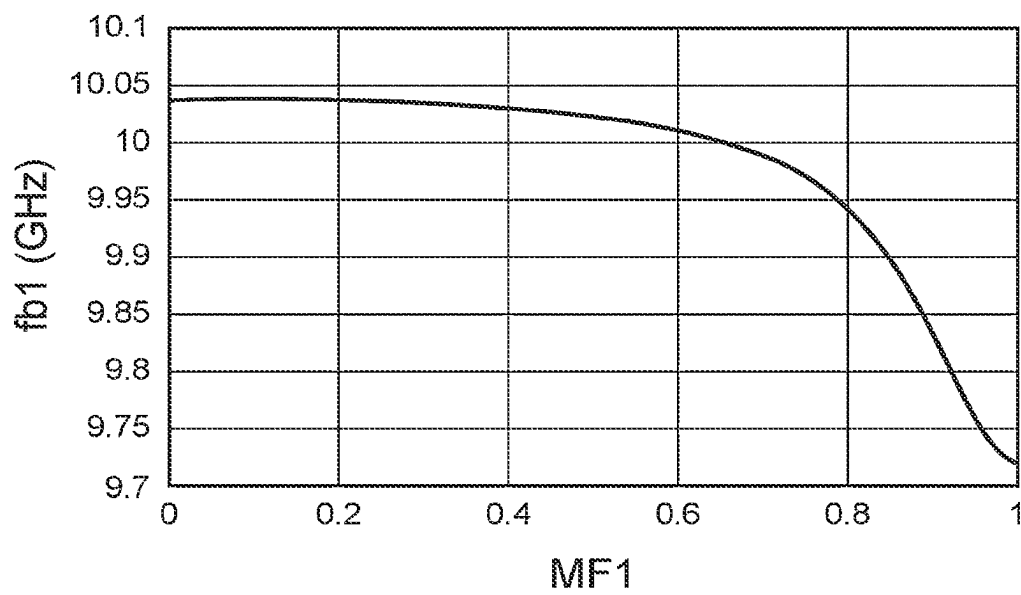


FIG. 10A

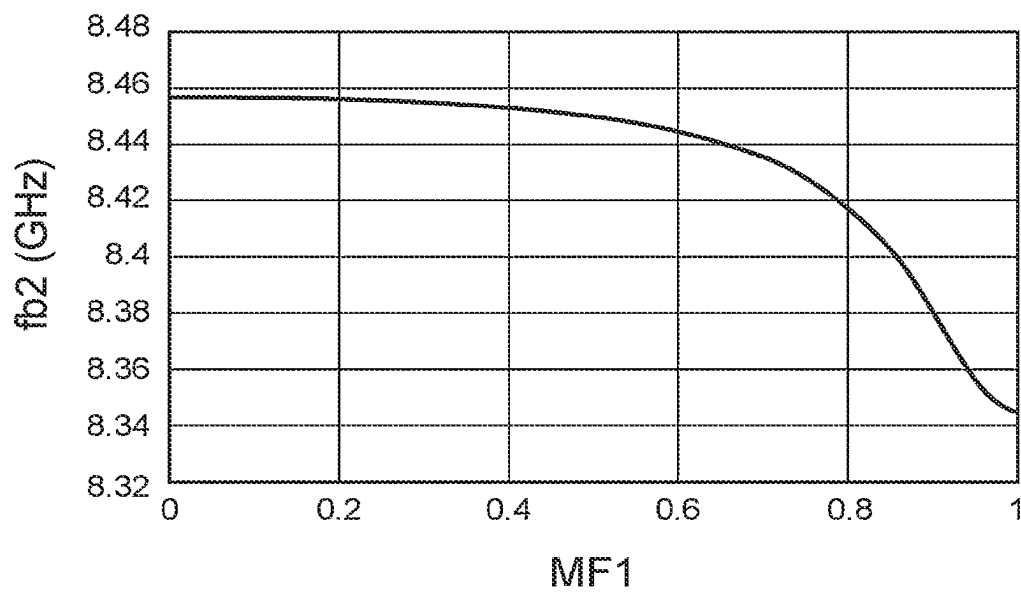


FIG. 10B

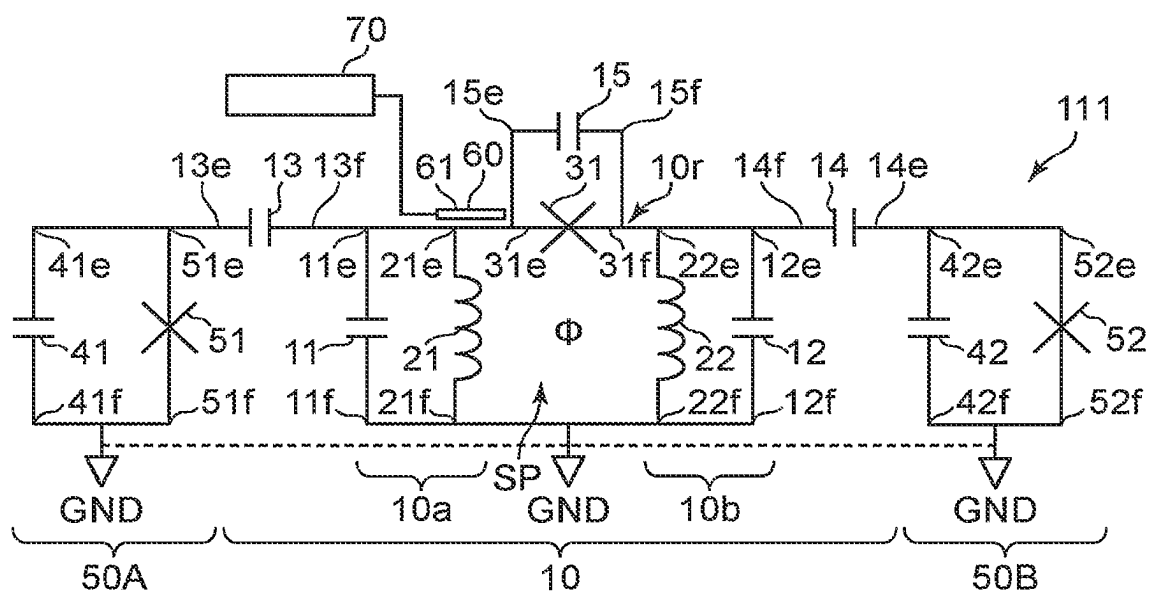


FIG. 11

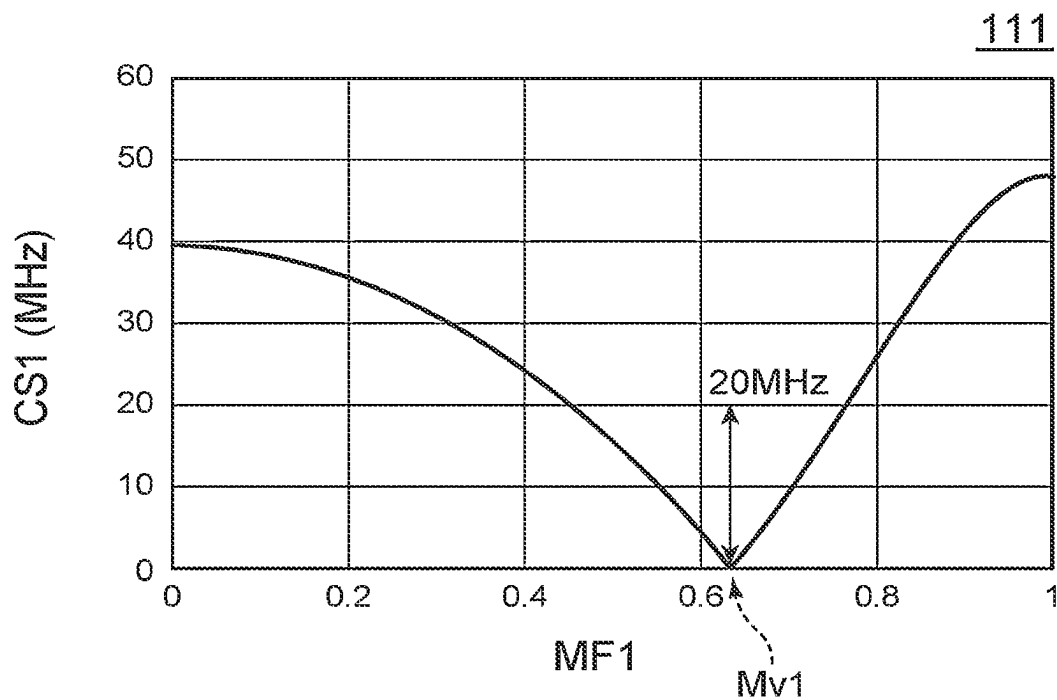


FIG. 12A

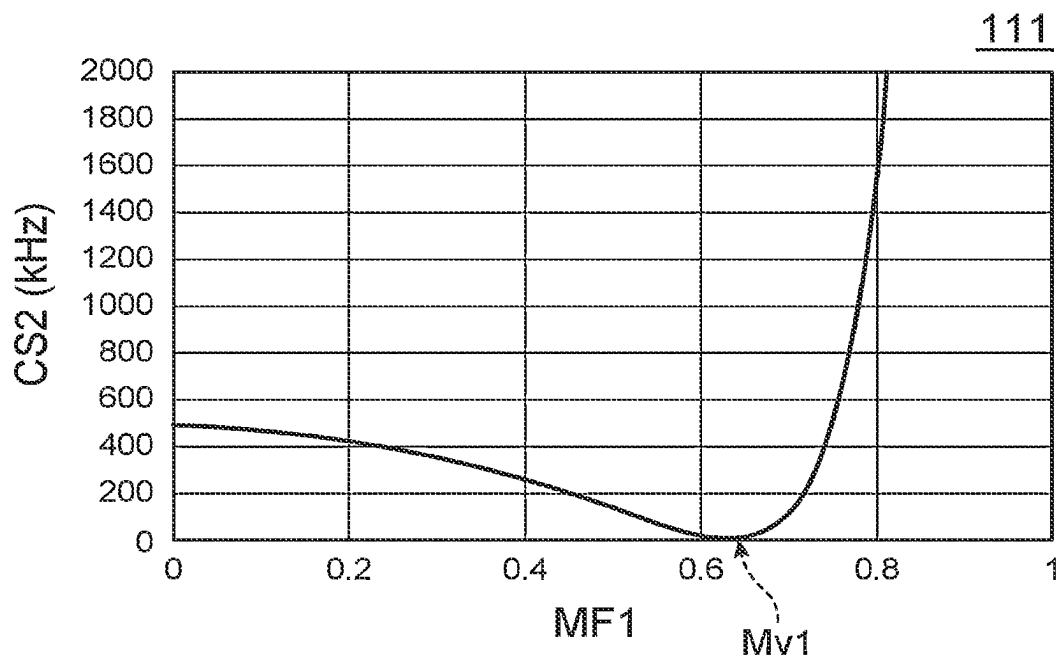


FIG. 12B

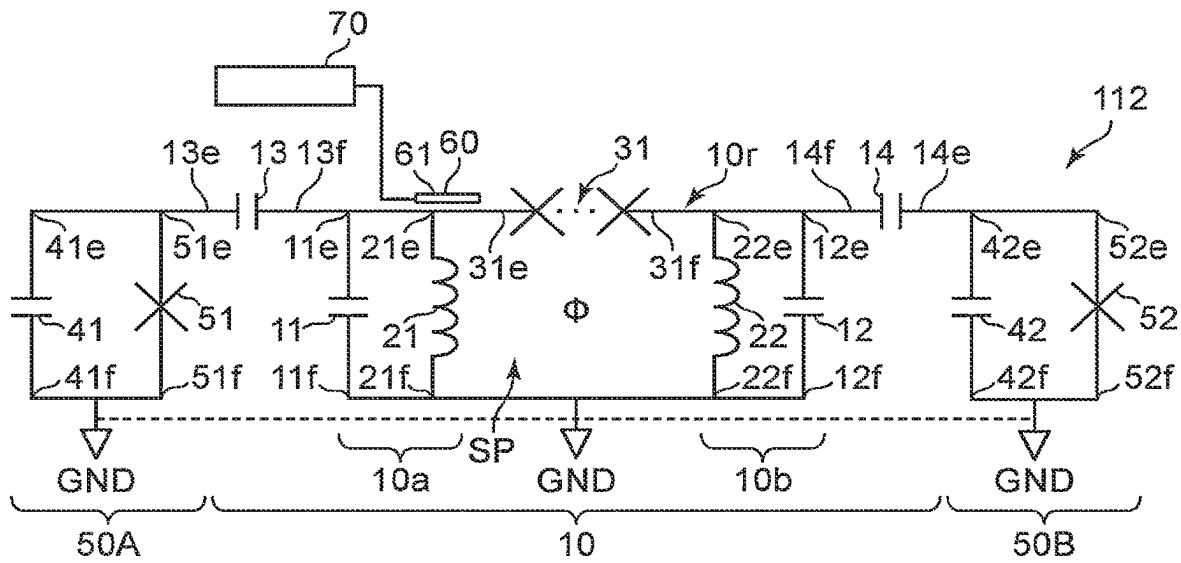


FIG. 13A

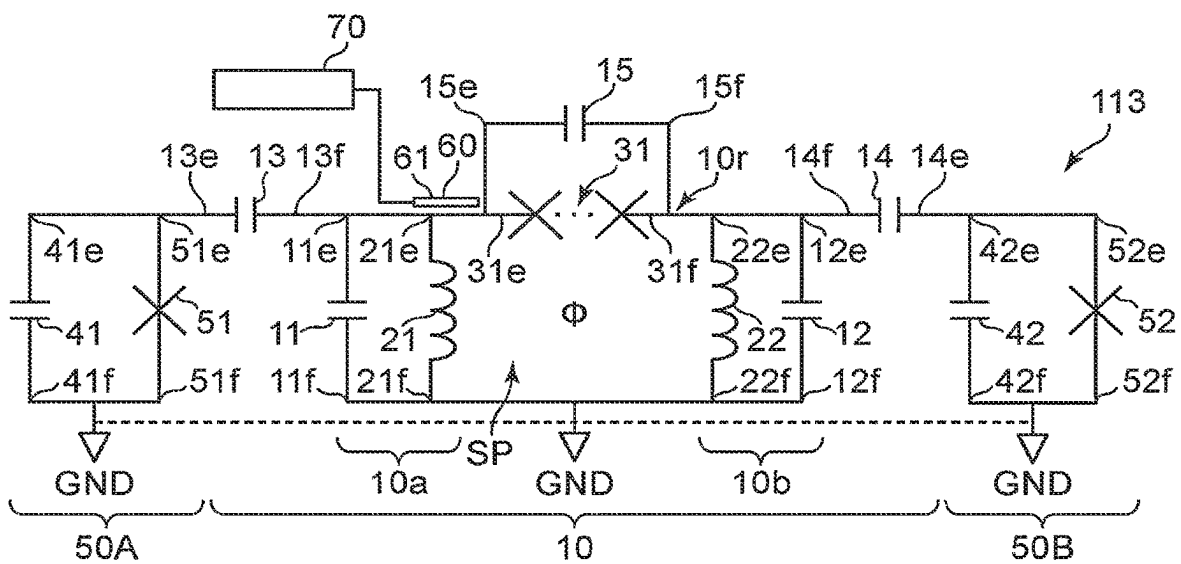


FIG. 13B

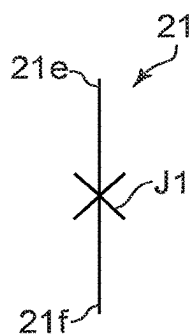


FIG. 14A

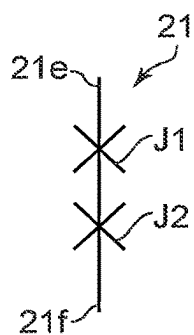


FIG. 14B

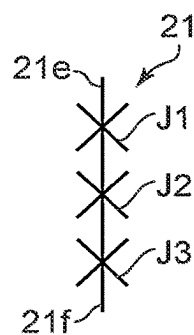


FIG. 14C

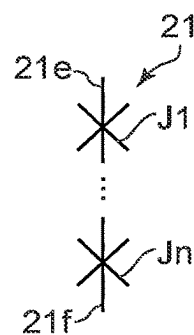


FIG. 14D

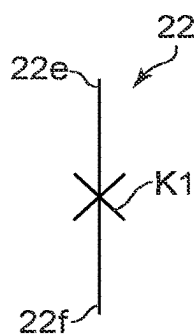


FIG. 14E

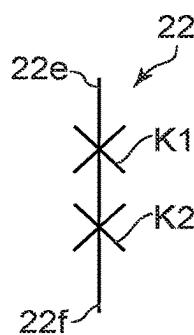


FIG. 14F

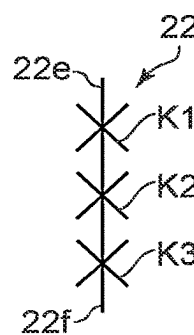


FIG. 14G

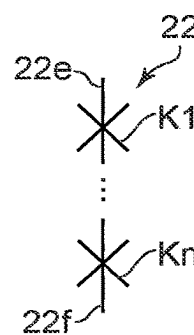


FIG. 14H

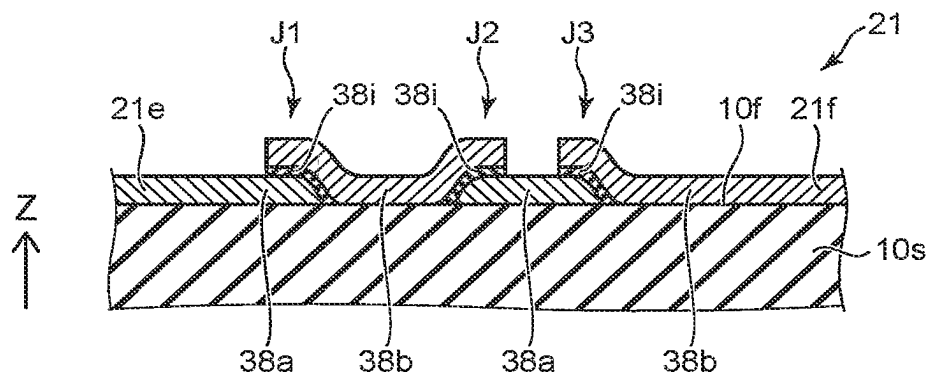


FIG. 15A

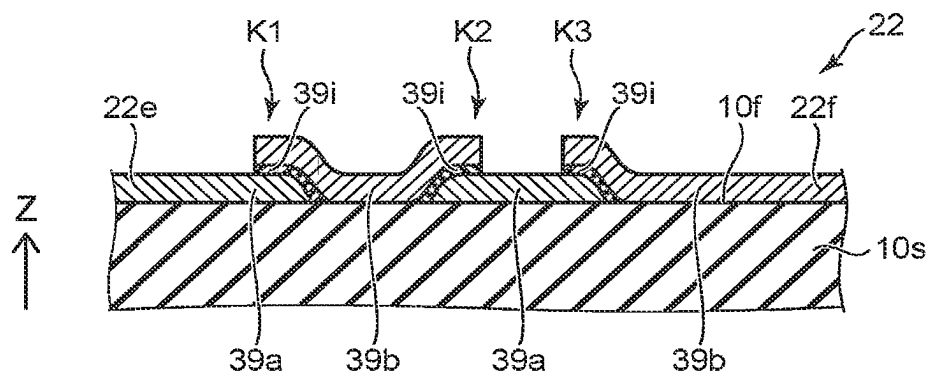


FIG. 15B

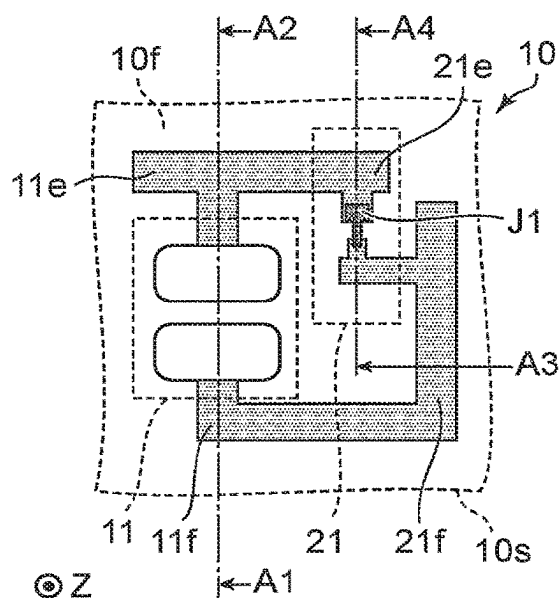


FIG. 16A

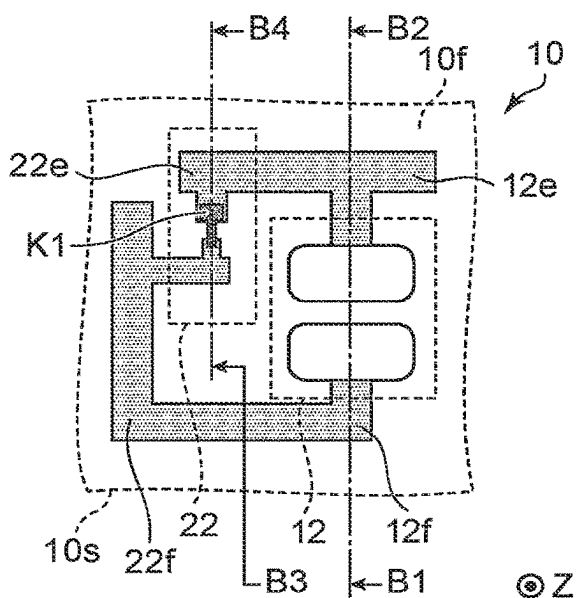


FIG. 16B

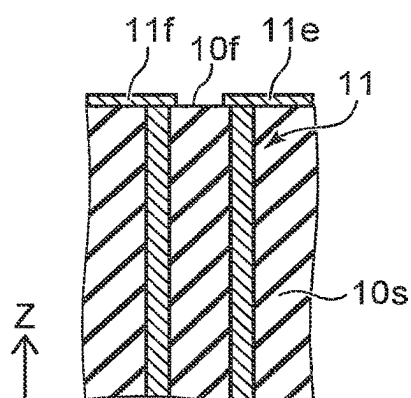


FIG. 16C

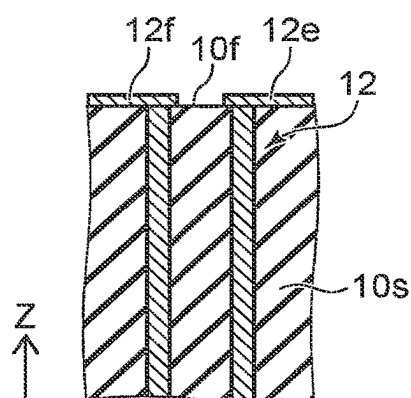


FIG. 16D

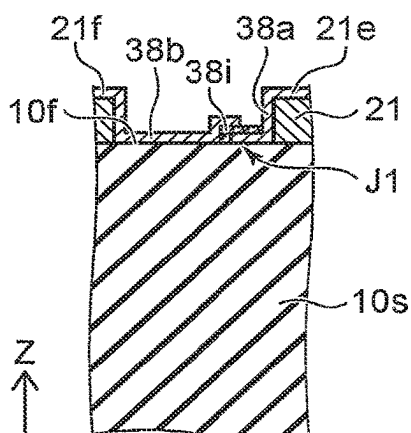


FIG. 16E

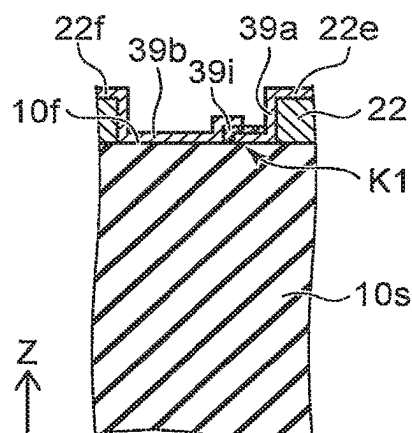


FIG. 16F

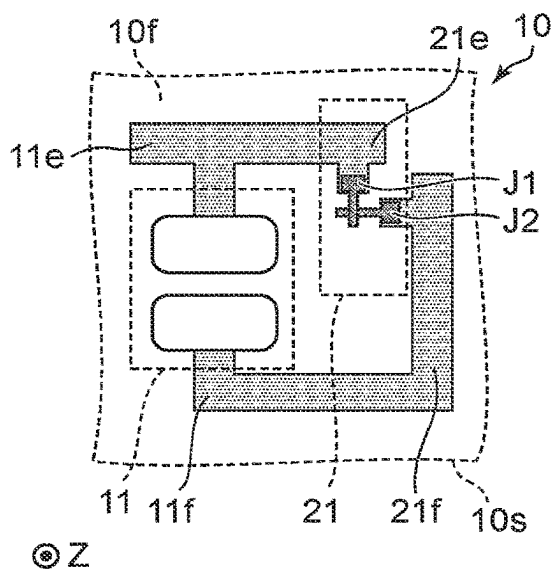


FIG. 17A

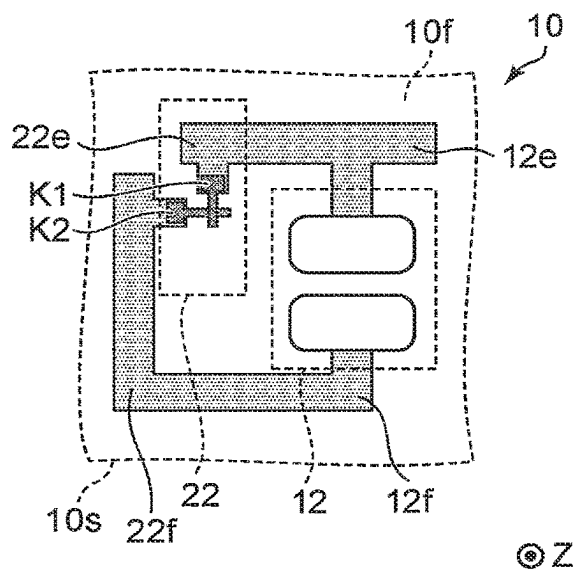


FIG. 17B

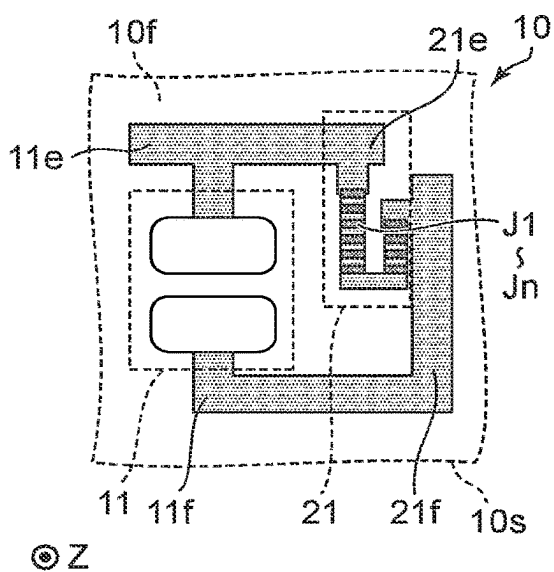


FIG. 18A

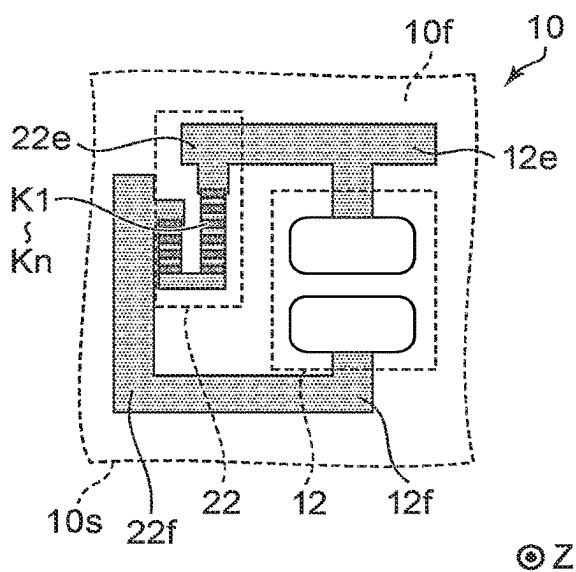


FIG. 18B

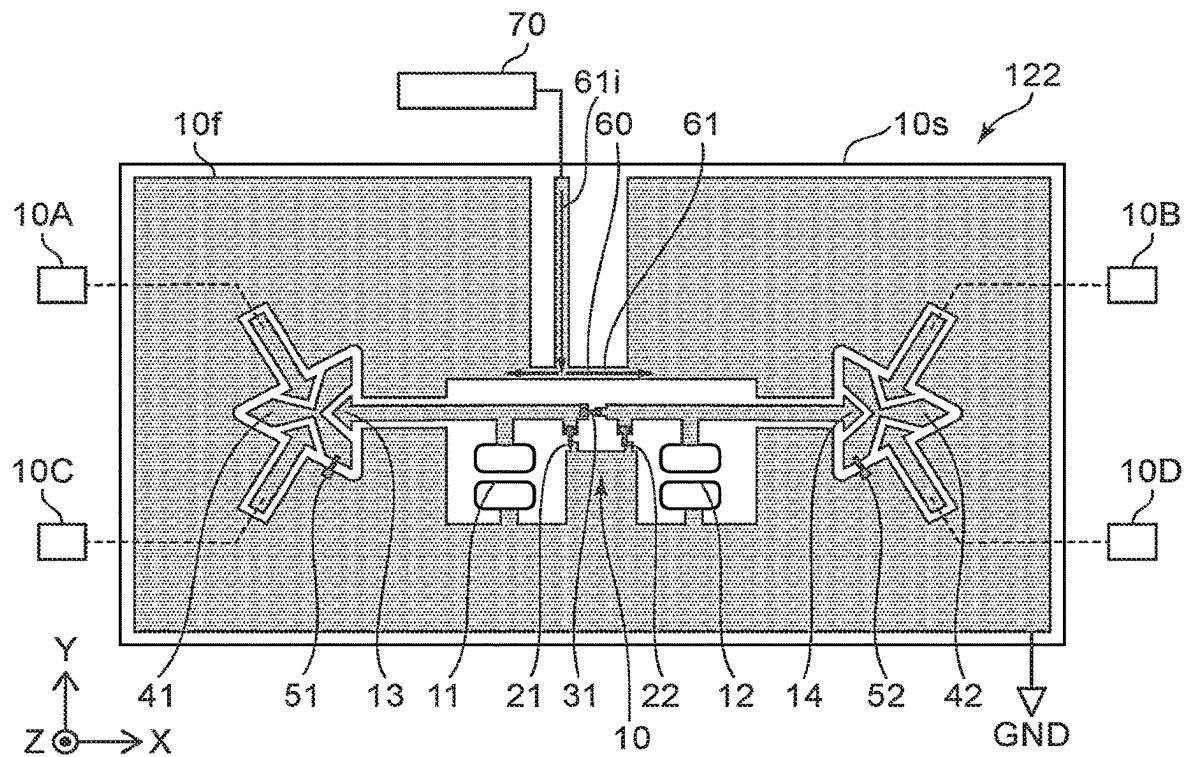


FIG. 21

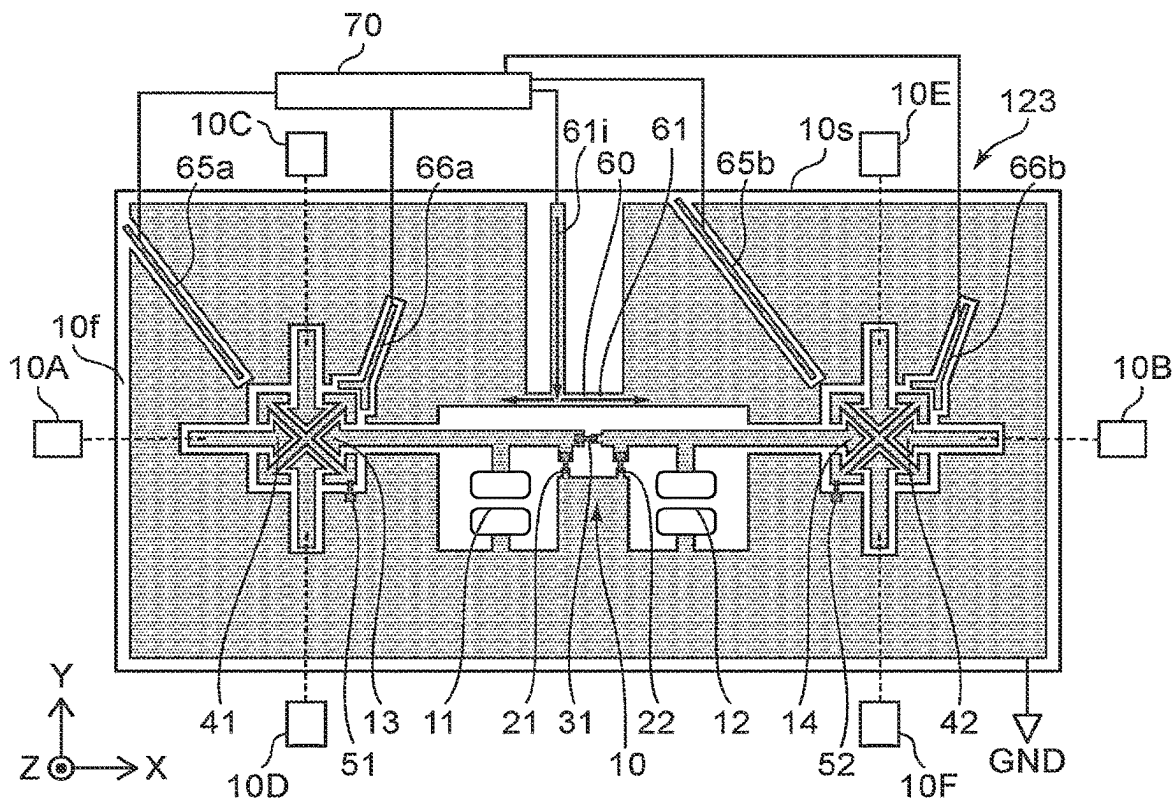


FIG. 22

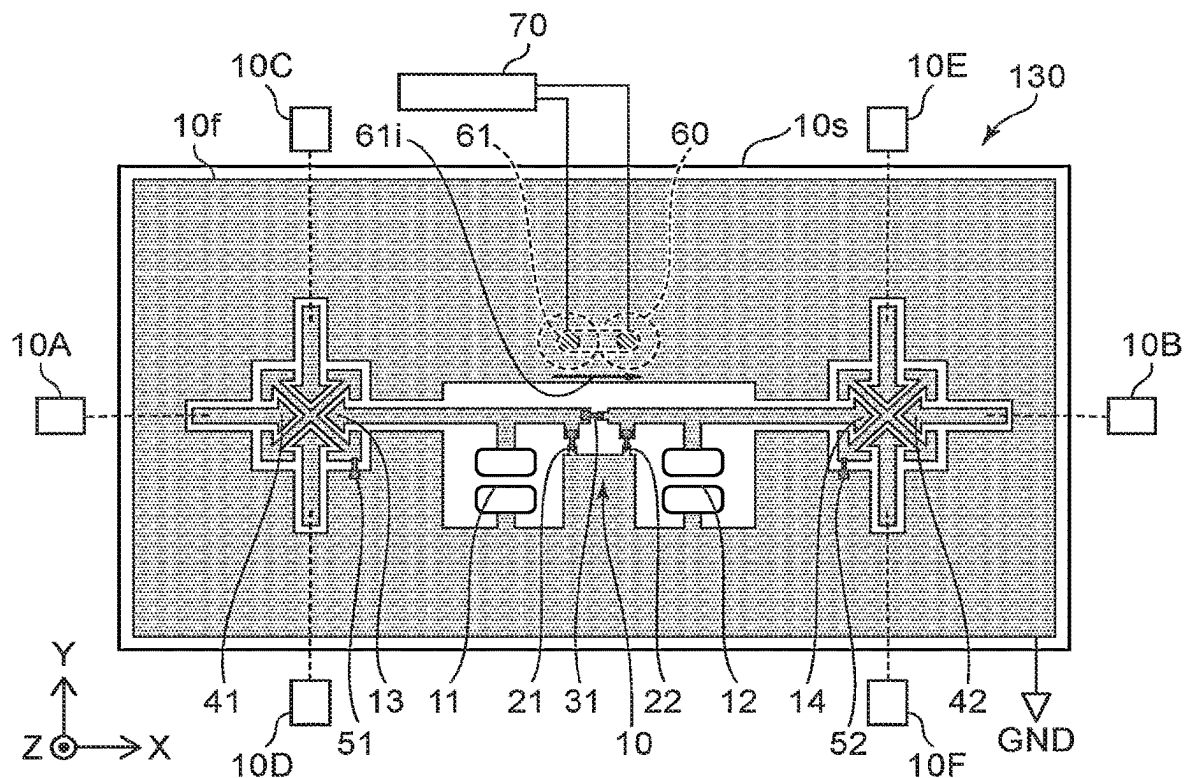


FIG. 23A

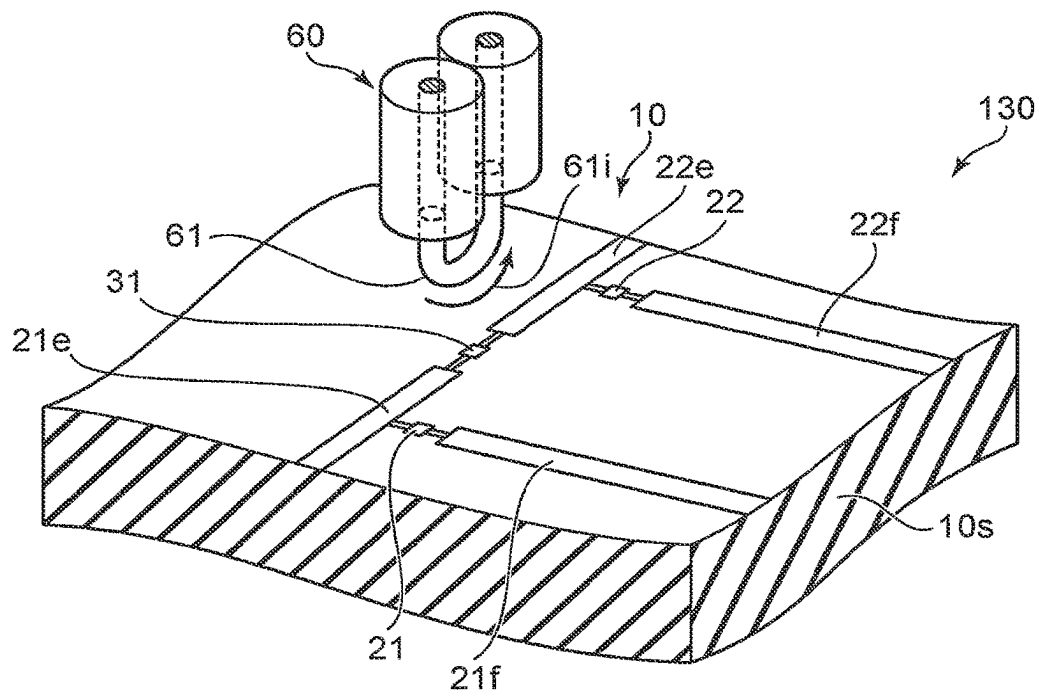
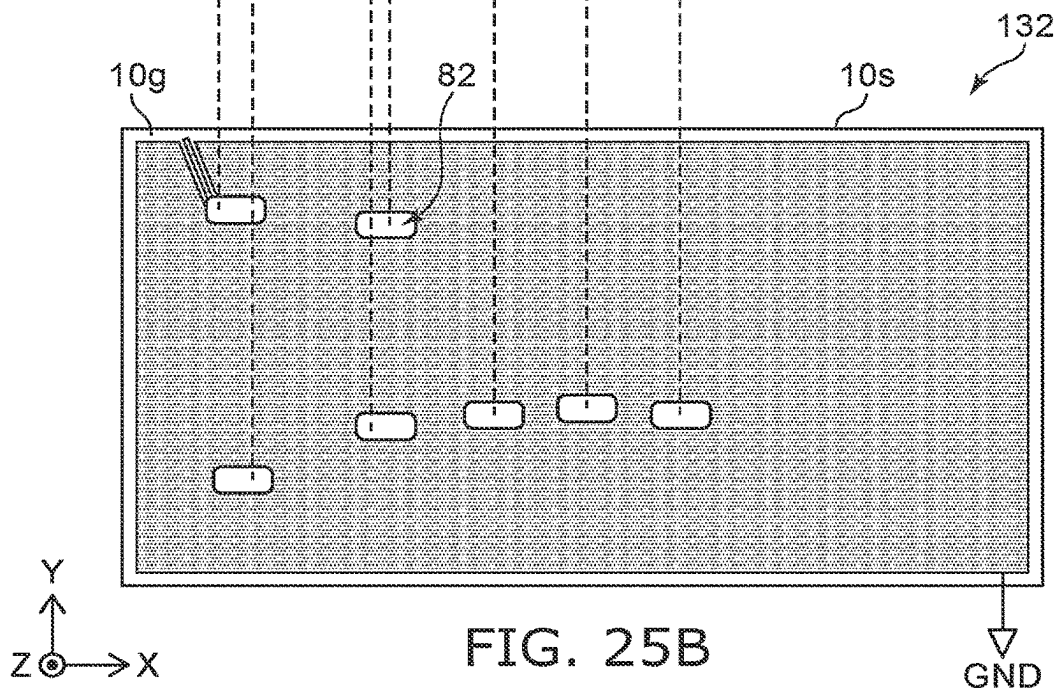
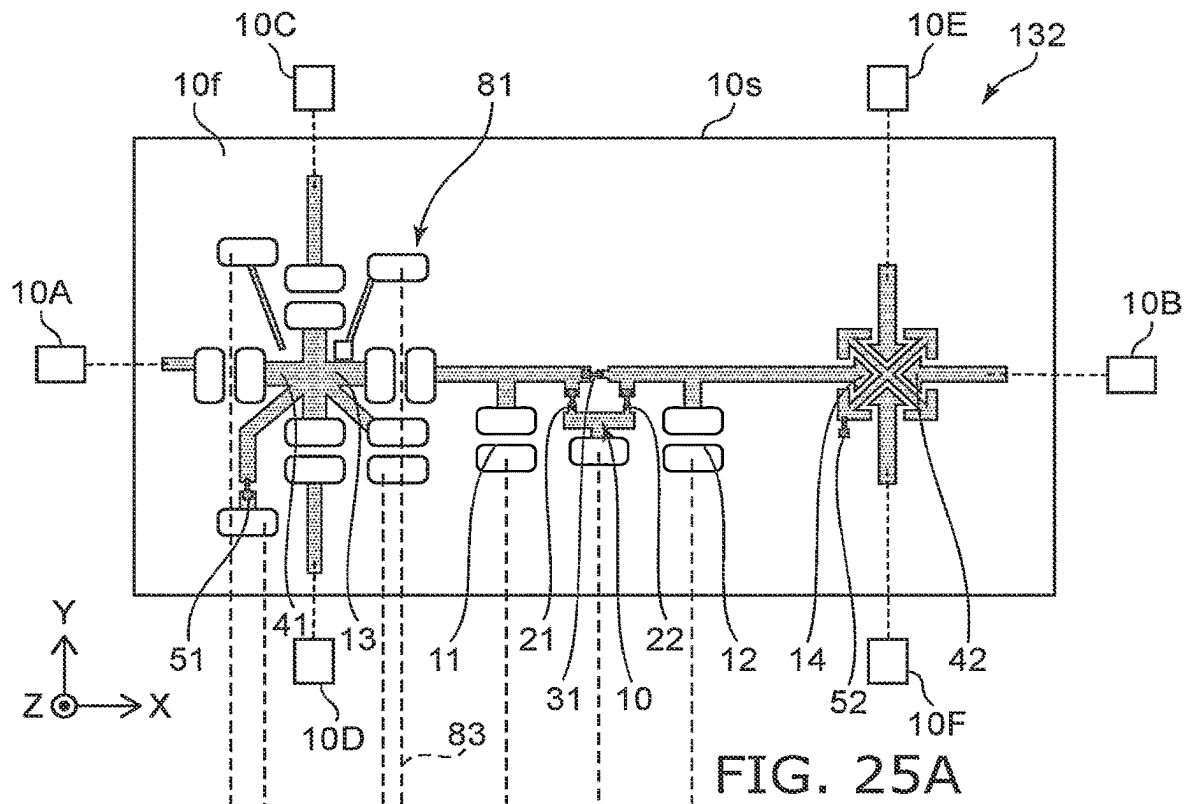


FIG. 23B

FIG. 24



COUPLER AND CALCULATING DEVICE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a divisional of U.S. patent application Ser. No. 18/348,676, filed Jul. 7, 2023, which is a divisional of U.S. patent application Ser. No. 17/931,515, filed Sep. 12, 2022, now U.S. Pat. No. 11,742,831, issued Aug. 29, 2023, which is a divisional of U.S. patent application Ser. No. 17/681,610, filed Feb. 25, 2022, now U.S. Pat. No. 11,476,836, issued Oct. 18, 2022, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-125184, filed on Jul. 30, 2021; the entire contents of which are incorporated herein by reference.

FIELD

Embodiments described herein relate generally to a coupler and a calculating device.

BACKGROUND

For example, a coupler is used in a calculating device that utilizes multiple nonlinear resonators. It is desirable to improve the controllability of the calculating device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view illustrating a coupler and a calculating device according to a first embodiment;

FIGS. 2A to 2C are schematic cross-sectional views illustrating portions of the coupler and the calculating device according to the first embodiment;

FIGS. 3A and 3B are schematic plan views illustrating portions of the coupler according to the first embodiment;

FIGS. 4A and 4B are graphs illustrating frequencies of couplers and calculating devices;

FIGS. 5A and 5B are graphs illustrating coupling strengths of the couplers and the calculating devices;

FIGS. 6A and 6B are graphs illustrating characteristics of the couplers and the calculating devices;

FIGS. 7A and 7B are graphs illustrating coupling strength and characteristic of a coupler and a calculating device of a second reference example;

FIGS. 8A and 8B are graphs illustrating coupling strength and characteristic of a coupler and a calculating device of a third reference example;

FIG. 9 is a graph illustrating the probability of the coupler and the calculating device according to the first embodiment;

FIGS. 10A and 10B are graphs illustrating resonant frequencies of the coupler and the calculating device according to the first embodiment;

FIG. 11 is a schematic view illustrating a coupler and a calculating device according to the first embodiment;

FIGS. 12A and 12B are graphs illustrating coupling strength and characteristic of a coupler and a calculating device according to the first embodiment;

FIGS. 13A and 13B are schematic views illustrating couplers and calculating devices according to the first embodiment;

FIGS. 14A to 14H are schematic views illustrating portions of a coupler according to a second embodiment;

FIGS. 15A and 15B are schematic cross-sectional views illustrating portions of the coupler according to the second embodiment;

FIGS. 16A and 16B are schematic plan views illustrating portions of a coupler of one example according to the second embodiment, and FIGS. 16C to 16F are schematic cross-sectional views illustrating the portions of the coupler of the one example according to the second embodiment;

FIGS. 17A and 17B are schematic plan views illustrating portions of a coupler of another example according to the second embodiment;

FIGS. 18A and 18B are schematic plan views illustrating portions of a coupler of another example according to the second embodiment;

FIG. 19 is a schematic plan view illustrating a coupler and a calculating device of another example according to the second embodiment;

FIG. 20 is a schematic plan view illustrating a coupler and a calculating device of another example according to the second embodiment;

FIG. 21 is a schematic plan view illustrating a coupler and a calculating device of another example according to the second embodiment;

FIG. 22 is a schematic plan view illustrating a coupler and a calculating device of another example according to the second embodiment;

FIG. 23A is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment, and FIG. 23B is a schematic perspective view illustrating a portion of the coupler and the calculating device according to the second embodiment;

FIG. 24 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment; and

FIGS. 25A and 25B are schematic plan views illustrating a coupler and a calculating device according to the second embodiment.

DETAILED DESCRIPTION

According to one embodiment, a coupler includes a first capacitor, a second capacitor, a third capacitor, a fourth capacitor, a first inductor, a second inductor, and a first Josephson junction. The first capacitor includes a first capacitor end portion and a first capacitor other-end portion. The first inductor includes a first inductor end portion and a first inductor other-end portion. The first inductor end portion is electrically connected with the first capacitor end portion. The second capacitor includes a second capacitor end portion and a second capacitor other-end portion. The second inductor includes a second inductor end portion and a second inductor other-end portion. The second inductor end portion is electrically connected with the second capacitor end portion. The second inductor other-end portion is electrically connected with the first capacitor other-end portion, the first inductor other-end portion, and the second capacitor other-end portion. The first Josephson junction includes a first Josephson junction end portion and a first Josephson junction other-end portion. The first Josephson junction end portion is electrically connected with the first capacitor end portion. The first Josephson junction other-end portion is electrically connected with the second capacitor other-end portion. A space is surrounded with the first inductor, the second inductor, and the first Josephson junction. The third capacitor includes a third capacitor end portion and a third capacitor other-end portion. The third capacitor other-end portion is electrically connected with the first capacitor end portion. The third capacitor end portion is electrically connectable with a first nonlinear resonator. The fourth capacitor includes a fourth capacitor end portion and a fourth

capacitor other-end portion. The fourth capacitor other-end portion is electrically connected with the second capacitor end portion. The fourth capacitor end portion is electrically connectable with a second nonlinear resonator.

Various embodiments are described below with reference to the accompanying drawings.

The drawings are schematic and conceptual; and the relationships between the thickness and width of portions, the proportions of sizes among portions, etc., are not necessarily the same as the actual values. The dimensions and proportions may be illustrated differently among drawings, even for identical portions.

In the specification and drawings, components similar to those described previously or illustrated in an antecedent drawing are marked with like reference numerals, and a detailed description is omitted as appropriate.

First Embodiment

FIG. 1 is a schematic view illustrating a coupler and a calculating device according to a first embodiment.

As shown in FIG. 1, the calculating device 110 according to the embodiment includes a coupler 10, a first nonlinear resonator 50A, and a second nonlinear resonator 50B. The coupler 10 couples the first nonlinear resonator 50A and the second nonlinear resonator 50B.

The coupler 10 includes a first capacitor 11, a second capacitor 12, a third capacitor 13, a fourth capacitor 14, a first inductor 21, a second inductor 22, and a first Josephson junction 31.

The first capacitor 11 includes a first capacitor end portion 11e and a first capacitor other-end portion 11f. The first inductor 21 includes a first inductor end portion 21e and a first inductor other-end portion 21f. The first inductor end portion 21e is electrically connected with the first capacitor end portion 11e.

The second capacitor 12 includes a second capacitor end portion 12e and a second capacitor other-end portion 12f. The second inductor 22 includes a second inductor end portion 22e and a second inductor other-end portion 22f. The second inductor end portion 22e is electrically connected with the second capacitor end portion 12e. The second inductor other-end portion 22f is electrically connected with the first capacitor other-end portion 11f, the first inductor other-end portion 21f, and the second capacitor other-end portion 12f. For example, the potentials of the first capacitor other-end portion 11f, the first inductor other-end portion 21f, the second capacitor other-end portion 12f, and the second inductor other-end portion 22f are set to a fixed potential (e.g., ground GND).

The first Josephson junction 31 includes a first Josephson junction end portion 31e and a first Josephson junction other-end portion 31f. The first Josephson junction end portion 31e is electrically connected with the first capacitor end portion 11e. The first Josephson junction end portion 31e also is electrically connected with the first inductor end portion 21e. The first Josephson junction other-end portion 31f is electrically connected with the second capacitor end portion 12e. The first Josephson junction other-end portion 31f also is electrically connected with the second inductor end portion 22e.

A space SP is surrounded with the first inductor 21, the second inductor 22, and the first Josephson junction 31. The space SP may be surrounded with the first capacitor 11, the second capacitor 12, and the first Josephson junction 31. For example, a loop 10r is formed from the first inductor 21, the second inductor 22, and the first Josephson junction 31. The

loop 10r surrounds the space SP. As described below, a magnetic flux Φ inside the space SP (inside the loop 10r) is controllable.

The third capacitor 13 includes a third capacitor end portion 13e and a third capacitor other-end portion 13f. The third capacitor other-end portion 13f is electrically connected with the first capacitor end portion 11e. The third capacitor other-end portion 13f also is electrically connected with the first inductor end portion 21e and the first Josephson junction end portion 31e. The third capacitor end portion 13e is connectable with the first nonlinear resonator 50A.

The fourth capacitor 14 includes a fourth capacitor end portion 14e and a fourth capacitor other-end portion 14f. The fourth capacitor other-end portion 14f is electrically connected with the second capacitor end portion 12e. The fourth capacitor other-end portion 14f also is electrically connected with the second inductor end portion 22e and the first Josephson junction other-end portion 31f. The fourth capacitor end portion 14e is connectable with the second nonlinear resonator 50B.

In the coupler 10, a first part 10a that includes the first capacitor 11 and the first inductor 21 corresponds to a first LC circuit. A second part 10b that includes the second capacitor 12 and the second inductor 22 corresponds to a second LC circuit. These LC circuits are connected by the first Josephson junction 31. The magnetic flux Φ of the space SP that is surrounded with these components can be modulated.

The coupler 10 includes multiple modes (e.g., two modes). According to the embodiment, by providing the aforementioned LC circuits in the coupler 10, compared to when only inductors are included, the resonant frequencies of the aforementioned multiple modes can be reduced. For example, the resonant frequencies of the multiple modes of the coupler 10 can easily approach the resonant frequency of the first nonlinear resonator 50A and the resonant frequency of the second nonlinear resonator 50B. A strong coupling strength is obtained thereby. According to the embodiment, the controllability can be improved.

As described below, the coupling strength can be modified by controlling the magnetic flux Φ . For example, the coupling strength can be substantially zero, and the coupling can be decoupled (switched off). As described below, a two-qubit gate operation can be quickly performed by the control of the coupler 10. A coupler and a calculating device can be provided in which the controllability can be improved.

As shown in FIG. 1, the first nonlinear resonator 50A includes, for example, a first nonlinear resonator Josephson junction 51 and a first resonator capacitor 41. An end portion 51e of the first nonlinear resonator Josephson junction 51 and an end portion 41e of the first resonator capacitor 41 are electrically connected with the third capacitor end portion 13e. Another end portion 51f of the first nonlinear resonator Josephson junction 51 and another end portion 41f of the first resonator capacitor 41 are electrically connected to each other. For example, the potentials of the other end portions 51f and 41f are set to a fixed potential (e.g., the ground GND). Accordingly, for example, the other end portion 51f of the first nonlinear resonator Josephson junction 51 and the other end portion 41f of the first resonator capacitor 41 may be electrically connected with the first capacitor other-end portion 11f.

The end portion 51e of the first nonlinear resonator Josephson junction 51 and the end portion 41e of the first resonator capacitor 41 are capacitively coupled with the first part 10a via the third capacitor 13.

The second nonlinear resonator **50B** includes, for example, a second nonlinear resonator Josephson junction **52** and a second resonator capacitor **42**. An end portion **52e** of the second nonlinear resonator Josephson junction **52** and an end portion **42e** of the second resonator capacitor **42** are electrically connected with the fourth capacitor end portion **14e**. Another end portion **52f** of the second nonlinear resonator Josephson junction **52** and another end portion **42f** of the second resonator capacitor **42** are electrically connected to each other. For example, the potentials of the other end portions **52f** and **42f** are set to a fixed potential (e.g., the ground GND). Accordingly, for example, the other end portion **52f** of the second nonlinear resonator Josephson junction **52** and the other end portion **42f** of the second resonator capacitor **42** may be electrically connected with the second capacitor other-end portion **12f**.

The end portion **52e** of the second nonlinear resonator Josephson junction **52** and the end portion **42e** of the second resonator capacitor **42** are capacitively coupled with the second part **10b** via the fourth capacitor **14**.

The first nonlinear resonator **50A** and the second nonlinear resonator **50B** function as two qubits. Among the multiple energy levels included in the first and second nonlinear resonators **50A** and **50B**, it is possible to use the two lowest energy levels as the two states of the qubit. The two lowest energy levels of the multiple energy levels correspond to a ground state and a first excited state. The aforementioned two states of the qubits correspond to computational basis states. For example, the resonant frequency of the first nonlinear resonator **50A** corresponds to a value of the energy difference between the two lowest energy states of the first nonlinear resonator **50A** converted into a frequency. For example, the resonant frequency of the second nonlinear resonator **50B** corresponds to a value of the energy difference between the two lowest energy states of the second nonlinear resonator **50B** converted into a frequency. The energy can be converted into a frequency corresponding to the energy by dividing by Planck's constant h .

As shown in FIG. 1, the coupler **10** may include a first conductive member **61**. The first conductive member **61** is configured to apply a magnetic field to the space SP (the loop **10r**). For example, the magnetic field is generated by a current supplied to the first conductive member **61**. The magnetic field that is generated is applied to the space SP (the loop **10r**). As described below, the coupling strength between the first nonlinear resonator **50A** and the second nonlinear resonator **50B** changes according to the magnetic flux Φ in the space SP (the loop **10r**) (the magnetic flux based on the magnetic field).

The first conductive member **61** is one example of a first magnetic field application part **60**. As shown in FIG. 1, the calculating device **110** may include a controller **70**. The coupler **10** (or the calculating device **110**) may include the first magnetic field application part **60**. The first magnetic field application part **60** is configured to apply a magnetic field to the space SP (the loop **10r**). The controller **70** is configured to change the magnetic flux Φ in the space SP (the loop **10r**) by controlling the first magnetic field application part **60**. When the first magnetic field application part **60** includes the first conductive member **61**, the controller **70** can change the magnetic flux Φ by modulating the current supplied to the first conductive member **61**.

FIGS. 2A to 2C are schematic cross-sectional views illustrating portions of the coupler and the calculating device according to the first embodiment.

As shown in FIG. 2A, for example, the first Josephson junction **31** is located on a first surface **10f** of a base body **10s**.

A direction perpendicular to the first surface **10f** is taken as a Z-axis direction. One direction perpendicular to the Z-axis direction is taken as an X-axis direction. A direction perpendicular to the Z-axis direction and the X-axis direction is taken as a Y-axis direction.

As shown in FIG. 2A, the first Josephson junction **31** includes, for example, a conductive film **35a**, a conductive film **35b**, and an insulating film **35i**. The insulating film **35i** is located between a portion of the conductive film **35a** and a portion of the conductive film **35b**.

As shown in FIG. 2B, the first nonlinear resonator Josephson junction **51** includes, for example, a conductive film **36a**, a conductive film **36b**, and an insulating film **36i**. The insulating film **36i** is located between a portion of the conductive film **36a** and a portion of the conductive film **36b**.

As shown in FIG. 2C, the second nonlinear resonator Josephson junction **52** includes, for example, a conductive film **37a**, a conductive film **37b**, and an insulating film **37i**. The insulating film **37i** is located between a portion of the conductive film **37a** and a portion of the conductive film **37b**. These conductive films are substantially along the X-Y plane.

These conductive films include, for example, at least one selected from the group consisting of Al, Nb, NbN, TiN, NbTiN, and Ta. These materials are superconducting materials. These insulating films include, for example, at least one selected from the group consisting of Al_2O_3 , Nb_2O_5 , NbO_2 , NbO, and AlN. The base body **10s** includes, for example, at least one selected from the group consisting of Si and sapphire. The base body **10s** is, for example, insulative.

FIGS. 3A and 3B are schematic plan views illustrating portions of the coupler according to the first embodiment.

In one example as shown in FIG. 3A, the first capacitor **11** and the first inductor **21** are located on the first surface **10f** of the base body **10s**. The first capacitor **11** is formed of two conductive layers **11L** that face each other. The first inductor **21** includes a first conductive layer **21L** that has a meandering structure.

In one example as shown in FIG. 3B, the second capacitor **12** and the second inductor **22** are located on the first surface **10f** of the base body **10s**. The second capacitor **12** is formed of two conductive layers **12L** that face each other. The second inductor **22** includes a second conductive layer **22L** that has a meandering structure.

The conductive layers described above include, for example, Al, Nb, NbN, TiN, NbTiN, Ta, etc.

In these examples, for example, the first inductor **21** and the second inductor **22** are based on kinetic inductors. As described below, the first inductor **21** and the second inductor **22** may include Josephson junctions.

An example of simulation results of characteristics of the coupler **10** (and the calculating device **110**) will now be described. The inductances of the first and second inductors **21** and **22** are 2.34 nH in the model of the following simulation. The critical current of the first Josephson junction **31** is 40 nA. The capacitances of the first and second capacitors **11** and **12** are 39 fF. The capacitances of the third and fourth capacitors **13** and **14** are 9.74 fF. The critical current of the first nonlinear resonator Josephson junction **51** is 70 nA. The critical current of the second nonlinear resonator Josephson junction **52** is 50 nA. The capacitances of the first and second resonator capacitors **41** and **42** are 39 fF. For example, the magnetic flux Φ of the space SP (the loop **10r**) can be changed by controlling the current supplied to the first conductive member **61**.

Characteristics related to the calculating device 110 according to the embodiment and characteristics related to a calculating device 119a of a first reference example will now be described. As described above, the coupler 10 includes LC circuits in the calculating device 110. In the calculating device 119a of the first reference example, the coupler 10 does not include the first capacitor 11 and the second capacitor 12. In other words, in the calculating device 119a, the first part 10a includes the first inductor 21 but does not include the first capacitor 11. In the calculating device 119a, the second part 10b includes the second inductor 22 but does not include the second capacitor 12. Otherwise, the configuration of the calculating device 119a is similar to the calculating device 110.

FIGS. 4A and 4B are graphs illustrating the frequencies of the couplers and the calculating devices.

FIG. 4A corresponds to the calculating device 110 according to the embodiment. FIG. 4B corresponds to the calculating device 119a of the first reference example. In these figures, the horizontal axis is a magnetic flux MF1 of the space SP (the loop 10r). The magnetic flux MF1 ($=2\Phi/\Phi_0$) is normalized by a reduced flux quantum Φ_0 and is dimensionless. In these figures, the vertical axis corresponds to a frequency fo1. These figures illustrate a resonant frequency fb1 of the first nonlinear resonator 50A and a resonant frequency fb2 of the second nonlinear resonator 50B. The first nonlinear resonator 50A corresponds to, for example, a first qubit. The second nonlinear resonator 50B corresponds to, for example, a second qubit. As described above, the resonant frequency of each nonlinear resonator corresponds to the value of the energy difference between the two lowest energy states of the nonlinear resonator converted into a frequency.

FIGS. 4A and 4B illustrate a frequency fc1 and a frequency fc2. The frequency fc1 corresponds to one frequency of the multiple modes (e.g., the two modes) of the coupler 10. The frequency fc2 corresponds to another frequency of the multiple modes (e.g., the two modes) of the coupler 10.

In the calculating device 110 according to the embodiment as shown in FIG. 4A, the frequency fc1 and the frequency fc2 change as the magnetic flux MF1 changes. In particular, the frequency fc2 greatly changes. In the example, the frequency fc1 and the frequency fc2 approach each other when the magnetic flux MF1 is about 0.66. In the example, the frequency fc2 is equal to the frequency fc1 at a first magnetic flux value Mv1. The first magnetic flux value Mv1 is about 0.66. In the example, the range of the frequencies fc1 and fc2 is not lower than 11 GHz and not more than 19 GHz.

As shown in FIG. 4A, the resonant frequency fb1 of the first nonlinear resonator 50A and the resonant frequency fb2 of the second nonlinear resonator 50B are substantially constant as the magnetic flux MF1 changes. In the example, the resonant frequency fb1 of the first nonlinear resonator 50A is about 10.0 GHz. The resonant frequency fb2 of the second nonlinear resonator 50B is about 8.4 GHz. An example of the amount of the change of the resonant frequencies fb1 and fb2 as the magnetic flux MF1 changes is described below.

Thus, according to the embodiment, the frequency fc1 and the frequency fc2 are relatively near the resonant frequencies fb1 and fb2. The coupler 10 includes multiple modes (at least two modes). In other words, the coupler 10 can resonate in multiple modes. At least two of the resonant frequencies (the frequency fc1 and the frequency fc2) of the multiple modes are higher than the resonant frequencies fb1 and fb2 and lower than the sum of the resonant frequency

fb1 and the resonant frequency fb2 at the vicinity of the first magnetic flux value Mv1 described above (the magnetic flux value at which the frequencies fc1 and fc2 are near each other). According to the embodiment, a state exists in which the resonant frequencies (the frequency fc1 and the frequency fc2) of the multiple modes are lower than the sum of the resonant frequency fb1 and the resonant frequency fb2.

According to the embodiment, the first part 10a and the second part 10b each include an inductor and a capacitor. Therefore, the resonant frequencies of the multiple modes of the coupler 10 can approach the resonant frequencies of the nonlinear resonators. For example, a strong coupling strength is obtained thereby. A coupler and a calculating device can be provided in which the controllability can be improved.

As shown in FIG. 4B, in the calculating device 119a of the first reference example, the resonant frequencies (the frequency fc1 and the frequency fc2) of the multiple modes of the coupler 10 are markedly higher than the resonant frequencies (the resonant frequency fb1 and the resonant frequency fb2) of the two nonlinear resonators. In the calculating device 119a as shown in FIG. 4B, the frequency fc1 and the frequency fc2 are higher than the sum of the resonant frequency fb1 and the resonant frequency fb2. Therefore, it is difficult to obtain a high coupling strength. In the first reference example, for example, the resonant frequencies of the multiple modes of the coupler 10 are not less than 3 times the resonant frequencies of the nonlinear resonators.

According to the embodiment, it is favorable for the capacitances of the first and second capacitors 11 and 12 to be large enough not to be negligible; for example, the capacitances of the first and second capacitors 11 and 12 are greater than 0.1 times the capacitances of the third and fourth capacitors 13 and 14. For example, the resonant frequencies (the frequency fc1 and the frequency fc2) of the multiple modes can be effectively reduced thereby.

According to the embodiment as shown in FIG. 4A, the resonant frequency fb1 of the first nonlinear resonator 50A and the resonant frequency fb2 of the second nonlinear resonator 50B are substantially constant with respect to the change of the magnetic flux MF1. For example, the frequencies of the two qubits substantially do not change. The frequencies of the two qubits are substantially fixed and are stable. A coupler and a calculating device can be provided in which the controllability can be improved.

FIGS. 5A and 5B are graphs illustrating coupling strengths of the couplers and the calculating devices.

FIG. 5A corresponds to the calculating device 110 according to the embodiment. FIG. 5B corresponds to the calculating device 119a of the first reference example described above. In these figures, the horizontal axis is the magnetic flux MF1 of the space SP (the loop 10r). The vertical axis is a coupling strength CS1.

As shown in FIG. 5A, the coupling strength CS1 changes as the magnetic flux MF1 changes. The coupling strength CS1 can be controlled by controlling the magnetic flux MF1. For example, the width of the change of the coupling strength CS1 is about 20 MHz. In other words, the coupling strength CS1 can be adjusted in a range of 20 MHz to 20 MHz.

Thus, the controller 70 can change the coupling strength CS1 between the first nonlinear resonator 50A and the second nonlinear resonator by controlling the magnetic flux Φ (the magnetic flux MF1) of the space SP.

In the example as shown in FIG. 5A, the coupling strength CS1 is substantially 0 when the magnetic flux MF1 is about

0.68. Thus, the controller 70 can substantially decouple the first nonlinear resonator 50A from the second nonlinear resonator 50B by controlling the magnetic flux Φ (the magnetic flux MF1) of the space SP. In other words, the coupling can be switched off. A coupler and a calculating device can be provided in which the controllability can be improved.

As shown in FIG. 5B, in the calculating device 119a of the first reference example as well, a condition exists at which the coupling strength CS1 is substantially 0. However, in the calculating device 119a, the magnetic flux MF1 when the coupling strength CS1 is not 0 is less than that of the calculating device 110. Thus, it is difficult to obtain a high coupling strength CS1 in the first reference example in which the capacitors are not included in the first and second parts 10a and 10b. According to the embodiment, a high coupling strength CS1 can be obtained, and the coupling can be switched off.

FIGS. 6A and 6B are graphs illustrating characteristics of the couplers and the calculating devices.

FIG. 6A corresponds to the calculating device 110 according to the embodiment. FIG. 6B corresponds to the calculating device 110a of the first reference example. In these figures, the horizontal axis is the magnetic flux MF1. In these figures, the vertical axis is a coupling strength CS2 related to residual coupling (so-called ZZ-coupling). ZZ-coupling corresponds to the state in which $fb1+fb2-fb3$ is nonzero due to the residual coupling, wherein the frequency $fb3$ corresponds to both the two qubits being in the “1 state”. The ZZ-coupling “shift” corresponds to the coupling strength CS2.

As described with reference to FIG. 4A, the coupling strength CS1 is substantially zero when the magnetic flux MF1 is about 0.68 (the first magnetic flux value Mv1). As shown in FIG. 6A, the coupling strength CS2 that is related to the residual coupling can be substantially zero when the magnetic flux MF1 is about 0.68. For example, robust zero ZZ-coupling is obtained.

On the other hand, in the calculating device 119a of the first reference example as shown in FIG. 6B, the coupling strength CS2 that is related to the residual coupling cannot be reduced and is not 0.

In the first reference example in which capacitors are not included in the first and second parts 10a and 10b, it is necessary to excessively increase the frequencies $fc1$ and $fc2$ to reduce the coupling strength CS2 related to the residual coupling; as a result, the coupling strength CS1 becomes markedly low. In the first reference example, it is difficult to obtain a high coupling strength CS1 while setting the coupling strength CS2 to zero.

A second reference example and a third reference example will now be described. In the second and third reference examples as well, capacitors are not included in the first and second parts 10a and 10b. In the second reference example, the values of the inductors are adjusted to obtain a high coupling strength CS1. In the third reference example, the values of the inductors are adjusted to reduce the coupling strength CS2 related to the residual coupling.

FIGS. 7A and 7B are graphs illustrating coupling strength and characteristic of the coupler and the calculating device of the second reference example.

These figures correspond to the calculating device 119b according to the second reference example. In the example of the second reference example, the inductances of the first and second inductors 21 and 22 are 4.1 nH. In FIGS. 7A and 7B, the horizontal axis is the magnetic flux MF1 of the space SP (the loop 10r). The vertical axis of FIG. 7A is the

coupling strength CS1. The vertical axis of FIG. 7B is the coupling strength CS2 that is related to the residual coupling.

In the second reference example as shown in FIG. 7A, a condition exists at which the coupling strength CS1 is substantially zero (the coupling is off). Compared to the first reference example, a high coupling strength CS1 is obtained in the second reference example. However, in the second reference example as shown in FIG. 7B, the coupling strength CS2 is large compared to that of the first reference example, and is not 0. Thus, in the second reference example, a low coupling strength CS2 cannot be obtained.

FIGS. 8A and 8B are graphs illustrating coupling strength and characteristic of the coupler and the calculating device of the third reference example.

These figures correspond to the calculating device 119c according to the third reference example. In the example of the third reference example, the inductances of the first and second inductors 21 and 22 are 1.63 nH. In FIGS. 8A and 8B, the horizontal axis is the magnetic flux MF1 of the space SP (the loop 10r). The vertical axis of FIG. 8A is the coupling strength CS1. The vertical axis of FIG. 8B is the coupling strength CS2 that is related to the residual coupling.

In the third reference example as shown in FIG. 8A, a condition exists at which the coupling strength CS1 is substantially zero (the coupling is off). In the third reference example, the coupling strength CS1 is markedly low. In the third reference example as shown in FIG. 8B, compared to the first reference example, the coupling strength CS2 becomes low but is substantially nonzero.

As described above, the first to third reference examples in which capacitors are not included in the first and second parts 10a and 10b cannot simultaneously obtain a coupling strength CS2 that is substantially zero, a high coupling strength CS1, and coupling that can be switched off.

According to the embodiment, capacitors are included in the first and second parts 10a and 10b. A coupling strength CS2 that is substantially zero, a high coupling strength CS1, and coupling that can be switched off are obtained thereby.

The element size increases when capacitors are included in the first and second parts 10a and 10b. Therefore, generally, the inclusion of capacitors in the first and second parts 10a and 10b is avoided.

Conversely, according to the embodiment as described above, capacitors are included in the first and second parts 10a and 10b. For example, a coupling strength CS2 that is substantially zero, a high coupling strength CS1, and coupling that can be switched off are obtained thereby. This is a special effect according to the embodiment. Such an effect is conventionally-unknown. This special effect was first discovered by fully quantum-mechanical analysis performed by the inventor of the application.

An example of characteristics related to a two-qubit gate operation will now be described. For example, the magnetic flux MF1 is set to a value (the first magnetic flux value Mv1 described above that is about 0.68) such that the coupling strength CS1 is zero when the two-qubit gate operation is not performed. The magnetic flux MF1 is modulated when performing the two-qubit gate operation. The frequency of the modulation is the difference between the resonant frequency $fb1$ and the resonant frequency $fb2$, and in the example, is about 1.55 GHz; the amplitude of the modulation of the simulation is 0.12. In other words, the magnetic flux MF1 oscillates with a sinusoidal waveform between 0.56 to 0.80.

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FIG. 9 is a graph illustrating a probability of the coupler and the calculating device according to the first embodiment. The horizontal axis of FIG. 9 is time t_m . The vertical axis is a probability P_1 .

FIG. 9 illustrates the probability P_1 of first to fourth states ST_1 to ST_4 . In the first state ST_1 , for example, the first nonlinear resonator **50A** is “1”, and the second nonlinear resonator **50B** is “0”. In the second state ST_2 , for example, the first nonlinear resonator **50A** is “0”, and the second nonlinear resonator **50B** is “1”. In the third state ST_3 , for example, the first nonlinear resonator **50A** is “1”, and the second nonlinear resonator **50B** is “1”. In the fourth state ST_4 , for example, the first nonlinear resonator **50A** is “0”, and the second nonlinear resonator **50B** is “0”.

As shown in FIG. 9, when the time t_m is 0, the probability P_1 of the first state ST_1 is 1, and the probability P_1 of the second state ST_2 is 0. The probability P_1 of the first state ST_1 and the probability P_1 of the second state ST_2 change as the time t_m elapses. In the example, the probability P_1 of the first state ST_1 is equal to the probability P_1 of the second state ST_2 when the time t_m is about 12 ns. The probability P_1 of the first state ST_1 drops below the probability P_1 of the second state ST_2 as the time t_m exceeds about 12 ns. In the example, the two-qubit gate operation can be completed at a time t_m of 12 ns.

According to the embodiment, a short gating time t_G is obtained. A two-qubit gate can be quickly performed. A coupler and a calculating device can be provided in which the controllability can be improved. The gating time t_G is, for example, not less than 10 ns and not more than about 20 ns.

Thus, according to the embodiment, the two-qubit gate operation for the first and second nonlinear resonators **50A** and **50B** is performed by modulating the magnetic flux MF_1 (the magnetic flux Φ) at an appropriate frequency. In the period in which the magnetic flux MF_1 is not modulated, the coupling remains off (the coupling strength CS_1 and the coupling strength CS_2 are substantially zero).

As shown in FIG. 9, the probability P_1 of the third state ST_3 and the probability P_1 of the fourth state ST_4 are substantially 0. A stable two-qubit gate is possible.

FIGS. 10A and 10B are graphs illustrating resonant frequencies of the coupler and the calculating device according to the first embodiment.

The horizontal axis of these figures is the magnetic flux MF_1 . The vertical axis of FIG. 10A is the resonant frequency fb_1 of the first nonlinear resonator **50A**. The vertical axis of FIG. 10B is the resonant frequency fb_2 of the second nonlinear resonator **50B**.

As shown in FIG. 10A, the resonant frequency fb_1 of the first nonlinear resonator **50A** changes at the vicinity of about 10 GHz when the magnetic flux MF_1 is changed from 0 to 1. The change amount (the change magnitude) of the resonant frequency fb_1 is about 300 MHz.

As shown in FIG. 10B, the resonant frequency fb_2 of the second nonlinear resonator **50B** changes at the vicinity of about 8.4 GHz when the magnetic flux MF_1 is changed from 0 to 1. The change amount (the change magnitude) of the resonant frequency fb_2 is about 100 MHz.

Thus, according to the embodiment, the changes of the resonant frequencies of the nonlinear resonators can be suppressed when the magnetic flux MF_1 is changed. The frequencies of the qubits are stable. The frequencies of the qubits can be substantially fixed. A coupler and a calculating device can be provided in which the controllability can be improved.

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According to the embodiment, for example, a high coupling strength CS_1 is obtained. For example, a fast two-qubit gate operation is possible. For example, the coupling of the multiple nonlinear resonators can be switched off while obtaining a high coupling strength CS_1 . The frequency change of the nonlinear resonators (the qubits) due to the magnetic flux is small; and the stability of the qubits is maintained. According to the embodiment, for example, a coupler and a calculating device can be provided in which the controllability can be improved.

For example, a fourth reference example may be considered in which two qubits of different frequencies are directly coupled, and a two-qubit gate is performed by irradiating a microwave. In the fourth reference example, the microwave of the frequency of one of the two qubits is irradiated on the other of the two qubits. In the fourth reference example, the characteristics are dependent on higher-order perturbation terms. High-speed gating is difficult in the fourth reference example.

For example, a fifth reference example may be considered in which two qubits of different frequencies are directly coupled, and a two-qubit gate is performed by irradiating a microwave. In the fifth reference example, the microwave that corresponds to the difference between the frequencies of the two qubits is irradiated. In the fifth reference example as well, the characteristics are dependent on higher-order perturbation terms. In the fifth reference example as well, high-speed gating is difficult.

Conversely, according to the embodiment, high-speed gating is possible by using the coupler **10** described above.

For example, the frequency of the “- mode” of the coupler **10** corresponds to the frequency fc_2 described above. According to the embodiment, the frequency of the “- mode” is adjusted using the magnetic flux Φ inside the loop **10r**. Thereby, the on and off switching of the coupling is possible. Coupling can be substantially completely off even when the difference between the frequencies of the two qubits is large. The coupling can be switched off even though the coupling strength is high. A higher speed also is possible. A coupler and a calculating device can be provided in which the controllability can be improved.

In one example according to the embodiment, for example, the capacitance of the first capacitor **11** is greater than 0.1 times the capacitance of the first resonator capacitor **41**. For example, the capacitance of the second capacitor **12** is greater than 0.1 times the capacitance of the second resonator capacitor **42**. The frequency of the coupler **10** is easily reduced thereby.

In one example according to the embodiment, for example, the capacitance of the third capacitor **13** is greater than 0.1 times the capacitance of the first resonator capacitor **41**. For example, the capacitance of the fourth capacitor **14** is greater than 0.1 times the capacitance of the second resonator capacitor **42**. The frequency of the coupler **10** is easily reduced thereby.

FIG. 11 is a schematic view illustrating a coupler and a calculating device according to the first embodiment.

In the calculating device **111** according to the embodiment as shown in FIG. 11, the coupler **10** includes a fifth capacitor **15**. Otherwise, the configuration of the calculating device **111** may be similar to the configuration of the calculating device **110**.

The fifth capacitor **15** includes a fifth capacitor end portion **15e** and a fifth capacitor other-end portion **15f**. The fifth capacitor end portion **15e** is electrically connected with the first capacitor end portion **11e**. The fifth capacitor end portion **15e** also is electrically connected with the first

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Josephson junction end portion **31e**, the first inductor end portion **21e**, and the third capacitor other-end portion **13f**. The fifth capacitor other-end portion **15f** is electrically connected with the second capacitor end portion **12e**. The fifth capacitor other-end portion **15f** also is electrically connected with the first Josephson junction other-end portion **31f**, the second inductor end portion **22e**, and the fourth capacitor other-end portion **14f**. For example, the fifth capacitor **15** makes it easier to adjust the coupling characteristics. For example, the position of the ZZ-coupling zero point can be adjusted. For example, by using the fifth capacitor **15**, a coupler and a calculating device that are robust with respect to magnetic flux fluctuation can be provided.

For example, the capacitance of the fifth capacitor **15** is less than the capacitance of the first capacitor **11** and less than the capacitance of the second capacitor **12**. The fifth capacitor **15** may be provided as necessary and may be omitted.

The first nonlinear resonator Josephson junction **51** and the second nonlinear resonator Josephson junction **52** may include parasitic capacitances. The parasitic capacitances are small (e.g., about 1 fF) compared to the capacitances of the first to fifth capacitors **11** to **15**, and are negligible.

FIGS. **12A** and **12B** are graphs illustrating coupling strength and characteristic of the coupler and the calculating device according to the first embodiment.

These figures correspond to the calculating device **111** according to the embodiment. In the example of the calculating device **111**, the capacitance of the fifth capacitor **15** is 1.6 fF. In FIGS. **12A** and **12B**, the horizontal axis is the magnetic flux MF1 of the space SP (the loop **10r**). The vertical axis of FIG. **12A** is the coupling strength CS1. The vertical axis of FIG. **12B** is the coupling strength CS2 that is related to the residual coupling.

In the calculating device **111** as shown in FIG. **12A**, a high coupling strength CS1 is obtained. A magnetic flux MF1 exists at which the coupling strength CS1 is substantially zero. The first magnetic flux value Mv1 is, for example, 0.62. As shown in FIG. **12B**, the coupling strength CS1 can be substantially zero. For example, the ZZ-coupling can be robustly set to zero by including the fifth capacitor **15**. It is easier to maintain the ZZ-coupling at substantially zero.

FIGS. **13A** and **13B** are schematic views illustrating couplers and calculating devices according to the first embodiment.

Multiple first Josephson junctions **31** are included in couplers **112** and **113** as shown in FIGS. **13A** and **13B**. The multiple first Josephson junctions **31** are connected in series. One end of the multiple first Josephson junctions **31** corresponds to the first Josephson junction end portion **31e**. The other end of the multiple first Josephson junctions **31** corresponds to the first Josephson junction other-end portion **31f**.

Second Embodiment

FIGS. **14A** to **14H** are schematic views illustrating a portion of a coupler according to a second embodiment.

As shown in FIG. **14A**, the first inductor **21** may include a first inductor Josephson junction (a Josephson junction **J1**, etc.). As shown in FIGS. **14B** and **14C**, the first inductor **21** may include multiple first inductor Josephson junctions (the Josephson junctions **J1** to **J3**, etc.). As shown in FIG. **14D**, the multiple first inductor Josephson junctions may include Josephson junctions **J1** to **Jn** (n being an integer that is not

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less than 2). The multiple first inductor Josephson junctions are electrically connected in series to each other.

As shown in FIG. **14E**, the second inductor **22** may include a second inductor Josephson junction (a Josephson junction **K1**, etc.). As shown in FIGS. **14F** and **14G**, the second inductor **22** may include multiple second inductor Josephson junctions (the Josephson junctions **K1** to **K3**, etc.). As shown in FIG. **14H**, the multiple second inductor Josephson junctions may include Josephson junctions **K1** to **Kn** (n being an integer that is not less than 2). The multiple second inductor Josephson junctions are electrically connected in series to each other.

FIGS. **15A** and **15B** are schematic cross-sectional views illustrating portions of the coupler according to the second embodiment.

As shown in FIG. **15A**, the first inductor **21** includes multiple first inductor Josephson junctions (the Josephson junctions **J1** to **J3**, etc.). The multiple first inductor Josephson junctions are located on the first surface **10f** of the base body **10s**. The multiple first inductor Josephson junctions (the Josephson junctions **J1** to **J3**, etc.) include a conductive film **38a**, a conductive film **38b**, and an insulating film **38i**. The insulating film **38i** is located between the conductive film **38a** and the conductive film **38b**.

As shown in FIG. **15B**, the second inductor **22** includes multiple second inductor Josephson junctions (the Josephson junctions **K1** to **K3**, etc.). The multiple second inductor Josephson junctions are located on the first surface **10f** of the base body **10s**. The multiple second inductor Josephson junctions (the Josephson junctions **K1** to **K3**, etc.) include a conductive film **39a**, a conductive film **39b**, and an insulating film **39i**. The insulating film **39i** is located between the conductive film **39a** and the conductive film **39b**.

The conductive film **38a**, the conductive film **38b**, the conductive film **39a**, and the conductive film **39b** may include the material included in the conductive film **35a**. The insulating film **38i** and the insulating film **39i** may include the material included in the insulating film **35i**.

FIGS. **16A** to **16F** are schematic views illustrating portions of the coupler according to the second embodiment.

FIGS. **16A** and **16B** are plan views. FIG. **16C** is a line A1-A2 cross-sectional view of FIG. **16A**. FIG. **16E** is a line A3-A4 cross-sectional view of FIG. **16A**. FIG. **16D** is a line B1-B2 cross-sectional view of FIG. **16B**. FIG. **16F** is a line B3-B4 cross-sectional view of FIG. **16B**.

As shown in FIG. **16C**, the first capacitor **11** may include two conductors extending along the Z-axis direction through the base body **10s**. As shown in FIG. **16D**, the second capacitor **12** may include two conductors extending along the Z-axis direction through the base body **10s**. These capacitors include, for example, vertical structures (or through-substrate-type structures). These capacitors may include, for example, TSV (through-silicon vias). The planar sizes of the capacitors can be small.

As shown in FIG. **16E**, the first inductor **21** includes the Josephson junction **J1**. In the example, the stacking direction of the conductive film **38a**, the insulating film **38i**, and the conductive film **38b** in the Josephson junction **J1** crosses the first surface **10f**. The coupling portion can be small.

As shown in FIG. **16F**, the second inductor **22** includes the Josephson junction **K1**. In the example, the stacking direction of the conductive film **39a**, the insulating film **39i**, and the conductive film **39b** in the Josephson junction **K1** crosses the first surface **10f**. The coupling portion can be small.

FIGS. **17A** and **17B** are schematic plan views illustrating portions of a coupler according to the second embodiment.

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As shown in FIG. 17A, the first inductor 21 includes the Josephson junctions J1 and J2. As shown in FIG. 17B, the second inductor 22 includes the Josephson junctions K1 and K2. In these Josephson junctions as well, the stacking direction crosses the first surface 10f.

FIGS. 18A and 18B are schematic plan views illustrating portions of a coupler according to the second embodiment.

As shown in FIG. 18A, the first inductor 21 includes multiple Josephson junctions (the Josephson junction J1 to the Josephson junction Jn). As shown in FIG. 18B, the second inductor 22 includes multiple Josephson junctions (the Josephson junctions K1 to Kn). In these Josephson junctions as well, the stacking direction crosses the first surface 10f.

FIG. 19 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment.

In the calculating device 120 according to the embodiment as shown in FIG. 19, the first conductive member 61 is included as the first magnetic field application part 60. At least a portion of the first conductive member 61 is along the X-Y plane. A magnetic flux control current 61i is supplied from the controller 70 to the first conductive member 61. The first resonator capacitor 41 (and the first nonlinear resonator Josephson junction 51) can be coupled with another coupler 10A. The second resonator capacitor 42 (and the second nonlinear resonator Josephson junction 52) can be coupled with another coupler 10B.

FIG. 20 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment.

In the calculating device 121 according to the embodiment as shown in FIG. 20, a portion that is included in the third capacitor 13 and a portion that is included in the first resonator capacitor 41 are oblique to the direction (in the example, the Y-axis direction) from the first capacitor other-end portion 11f toward the first capacitor end portion 11e. A portion that is included in the fourth capacitor 14 and a portion that is included in the second resonator capacitor 42 are oblique to the direction (in the example, the Y-axis direction) from the second capacitor other-end portion 12f toward the second capacitor end portion 12e. The first resonator capacitor 41 (and the first nonlinear resonator Josephson junction 51) can be coupled with the other couplers 10A, 10C, and 10D. The second resonator capacitor 42 (and the second nonlinear resonator Josephson junction 52) can be coupled with the other couplers 10B, 10E, and 10F.

FIG. 21 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment.

In the calculating device 122 according to the embodiment as shown in FIG. 21, the first resonator capacitor 41 (and the first nonlinear resonator Josephson junction 51) can be coupled with the other couplers 10A and 10C. The second resonator capacitor 42 (and the second nonlinear resonator Josephson junction 52) can be coupled with the other couplers 10B and 10D.

FIG. 22 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment.

As shown in FIG. 22, the calculating device 123 according to the embodiment may include a first excitation conductive portion 65a and a second excitation conductive portion 65b. The controller 70 can excite the first nonlinear resonator 50A via the first excitation conductive portion 65a. The controller 70 can excite the second nonlinear resonator 50B via the second excitation conductive portion 65b. For example, a signal that includes an AC component is supplied to the first and second excitation conductive portions 65a and 65b.

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As shown in FIG. 22, the calculating device 123 may include a first read resonator 66a and a second read resonator 66b. The controller 70 is configured to detect the state of the first nonlinear resonator 50A via the first read resonator 66a.

The controller 70 is configured to detect the state of the second nonlinear resonator 50B via the second read resonator 66b. For example, the resonant frequency fb1 of the first nonlinear resonator 50A can be identified using a reflection spectrum measurement via the first excitation conductive portion 65a or the first read resonator 66a. For example, the resonant frequency fb2 of the second nonlinear resonator 50B can be identified using a reflection spectrum measurement via the second excitation conductive portion 65b or the second read resonator 66b.

FIGS. 23A and 23B are schematic views illustrating a coupler and a calculating device according to the second embodiment.

FIG. 23A is a plan view. FIG. 23B is an enlarged perspective view of a portion of the coupler and the calculating device.

In the calculating device 130 as shown in FIGS. 23A and 23B, the first conductive member 61 is included as the first magnetic field application part 60. For example, the direction from the coupler 10 toward at least a portion of the first conductive member 61 is along the Z-axis direction (a direction crossing the first surface 10f). The first conductive member 61 is, for example, a coaxial line. The magnetic flux control current 61i includes a component in a direction that crosses the Z-axis direction (a direction along the X-Y plane) where the first conductive member 61 approaches the coupler 10.

FIG. 24 is a schematic plan view illustrating a coupler and a calculating device according to the second embodiment.

In the calculating device 131 as shown in FIG. 24, the first resonator capacitor 41 has a vertical structure (referring to FIGS. 16C and 16D). The vertical structure includes two conductors extending along the Z-axis direction through the base body 10s.

FIGS. 25A and 25B are schematic plan views illustrating a coupler and a calculating device according to the second embodiment.

FIG. 25A illustrates a conductive pattern 81 provided at the first surface 10f of the base body 10s. FIG. 25B illustrates a conductive pattern 82 provided at a second surface 10g of the base body 10s. The second surface 10g is the surface at the side opposite to the first surface 10f. In the calculating device 132 as shown in FIGS. 25A and 25B, a portion of the conductive pattern 81 is electrically connected with the conductive pattern 82 by a conductive member 83 that is along the Z-axis direction through the base body 10s. For example, an electrical connection may be performed to the outside via the conductive pattern 82. The conductive member 83 is, for example, a through-substrate-type connection member. According to the embodiment, a portion of a conductive member (the ground GND of a fixed potential or a circuit element) may be located at one of the first surface 10f or the second surface 10g of the base body 10s; another portion of the conductive member may be located at the other of the first surface 10f or the second surface 10g; and the portion and the other portion may be connected by a through-substrate-type connection member. According to the embodiment, connections may be performed by capacitive coupling in a flip-chip structure, etc.

Other than the description of the second embodiment described above, the configurations described in reference to the first embodiment are applicable to the second embodiment within the extent of technical feasibility.

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Characteristics of the coupler **10** and the calculating device **110** according to embodiments will now be described.

The Lagrangian of a system that includes the coupler **10**, the first nonlinear resonator **50A** coupled with the coupler **10**, and the second nonlinear resonator **50B** coupled with the coupler **10** is represented by the following first formula.

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_c + \mathcal{L}_I \quad (1)$$

The left side of the first formula is the Lagrangian of the system that includes the coupler **10**, the first nonlinear resonator **50A** coupled with the coupler **10**, and the second nonlinear resonator **50B** coupled with the coupler **10**.

The first term of the right side of the first formula is the Lagrangian of the first nonlinear resonator **50A**. The second term of the right side of the first formula is the Lagrangian of the second nonlinear resonator **50B**. The third term of the right side of the first formula is the Lagrangian of the coupler **10**. The fourth term of the right side of the first formula is the Lagrangian of the interaction between the coupler **10**, the first nonlinear resonator **50A**, and the second nonlinear resonator **50B**.

The Lagrangian of the first nonlinear resonator **50A** is represented by the following second formula. In the second formula, “ C_1 ” is the capacitance of the first resonator capacitor **41**.

$$\mathcal{L}_1 = \frac{C_1}{2} \dot{\phi}_1^2 + \phi_0 I_{c1} \cos \theta_1 \quad (2)$$

The Lagrangian of the second nonlinear resonator **50B** is represented by the following third formula. In the third formula, “ C_2 ” is the capacitance of the second resonator capacitor **42**.

$$\mathcal{L}_2 = \frac{C_2}{2} \dot{\phi}_2^2 + \phi_0 I_{c2} \cos \theta_2 \quad (3)$$

The Lagrangian of the interaction between the coupler **10**, the first nonlinear resonator **50A**, and the second nonlinear resonator **50B** is represented by the following fourth formula. In the fourth formula, “ C_c ” is the capacitance of each of the third and fourth capacitors **13** and **14**.

$$\mathcal{L}_I = \frac{C_c}{2} (\dot{\phi}_1 - \dot{\phi}_{c1})^2 + \frac{C_c}{2} (\dot{\phi}_2 - \dot{\phi}_{c2})^2 = \frac{C_c}{2} \left(\dot{\phi}_1 - \frac{\dot{\phi}_{c+} + \dot{\phi}_{c1}}{2} \right)^2 + \frac{C_c}{2} \left(\dot{\phi}_2 - \frac{\dot{\phi}_{c+} - \dot{\phi}_{c1}}{2} \right)^2 \quad (4)$$

The Lagrangian of the coupler **10** is represented by the following fifth formula. In the fifth formula, “ C ” is the capacitance of each of the first and second capacitors **11** and **12**.

$$\mathcal{L}_c = \frac{C}{2} \dot{\phi}_{c1}^2 - \frac{\phi_{c1}^2}{2L} + \frac{C}{2} \dot{\phi}_{c2}^2 - \frac{\phi_{c2}^2}{2L} + \frac{C_I}{2} (\dot{\phi}_{c1} - \dot{\phi}_{c2} + \Phi)^2 + \phi_0 I_c \cos \left(\theta_{c1} - \theta_{c2} + \frac{\Phi}{\phi_0} \right) = \quad (5)$$

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-continued

$$\frac{C}{4} \dot{\phi}_c^2 + \frac{\phi_c^2}{4L} + \frac{C}{4} \dot{\phi}_{c-}^2 - \frac{\phi_{c-}^2}{4L} + \frac{C_I}{2} (\dot{\phi}_{c-} + \Phi)^2 + \phi_0 I_c \cos \left(\theta_{c-} + \frac{\Phi}{\phi_0} \right)$$

Here, Φ is the magnetic flux operator. Φ has the relationship with a phase difference θ represented by the following sixth formula.

$$\phi = \phi_0 \theta \quad (6)$$

A magnetic flux operator Φ_{c+} for the “+ mode” of the coupler **10** is represented by the following seventh formula.

$$\phi_{c+} \equiv \phi_{c1} + \phi_{c2} \quad (7)$$

A magnetic flux operator Φ_{c-} for the “− mode” of the coupler **10** is represented by the following eighth formula.

$$\phi_{c-} \equiv \phi_{c1} - \phi_{c2} \quad (8)$$

In the seventh and eighth formulas, ϕ_{c1} is the magnetic flux operator for the first part **10a**. In the eighth and ninth formulas, ϕ_{c2} is the magnetic flux operator for the second part **10b**.

The signs of the first and second terms are interchanged on the right side of the fourth formula recited above. The coupling between the qubits via the + modes cancels.

The first term and the second term on the right side of the fifth formula recited above correspond to the “+ mode”. The third term to the sixth term on the right side of the fifth formula correspond to the “− mode”. The “+ mode” corresponds to an LC resonator. The “− mode” corresponds to a fluxonium qubit. The frequency is changeable due to the existence of the fluxonium qubit.

Thus, according to the embodiment, the coupler **10** simultaneously includes the two modes of the “+ mode” and the “− mode”. A changeable frequency is obtained by utilizing the “− mode”.

To simplify the above description, the case is described where the first capacitor **11** and the second capacitor **12** are the same (C). For simplification, the case is described where the capacitances of the third and fourth capacitors **13** and **14** are the same (C_c). According to the embodiment, the capacitance of the first capacitor **11** may be different from the capacitance of the second capacitor **12**. According to the embodiment, the capacitance of the third capacitor **13** may be different from the capacitance of the fourth capacitor **14**.

Embodiments may include the following configurations (e.g., technological proposals).

Configuration 1

A coupler, comprising:

- a first capacitor including a first capacitor end portion and a first capacitor other-end portion;
- a first inductor including a first inductor end portion electrically connected with the first capacitor end portion, and
- a first inductor other-end portion;
- a second capacitor including a second capacitor end portion and a second capacitor other-end portion;

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- a second inductor including
 - a second inductor end portion electrically connected with the second capacitor end portion, and
 - a second inductor other-end portion electrically connected with the first capacitor other-end portion, the first inductor other-end portion, and the second capacitor other-end portion;
- a first Josephson junction including
 - a first Josephson junction end portion electrically connected with the first capacitor end portion, and
 - a first Josephson junction other-end portion electrically connected with the second capacitor end portion, a space being surrounded with the first inductor, the second inductor, and the first Josephson junction;
- a third capacitor including
 - a third capacitor end portion electrically connectable with a first nonlinear resonator, and
 - a third capacitor other-end portion electrically connected with the first capacitor end portion; and
- a fourth capacitor including
 - a fourth capacitor end portion electrically connectable with a second nonlinear resonator, and
 - a fourth capacitor other-end portion electrically connected with the second capacitor end portion.

Configuration 2

The coupler according to Configuration 1, wherein the first inductor includes a first conductive layer having a meandering structure, and the second inductor includes a second conductive layer having a meandering structure.

Configuration 3

The coupler according to Configuration 1, wherein the first inductor includes a first inductor Josephson junction, and the second inductor includes a second inductor Josephson junction.

Configuration 4

The coupler according to Configuration 1, wherein the first inductor includes a plurality of first inductor Josephson junctions, the plurality of first inductor Josephson junctions is electrically connected in series to each other, the second inductor includes a plurality of second inductor Josephson junctions, and the plurality of second inductor Josephson junctions is electrically connected in series to each other.

Configuration 5

- The coupler according to any one of Configurations 1 to 4, further comprising:
- a fifth capacitor including a fifth capacitor end portion and a fifth capacitor other-end portion,
 - the fifth capacitor end portion being electrically connected with the first capacitor end portion,
 - the fifth capacitor other-end portion being electrically connected with the second capacitor end portion.

Configuration 6

The coupler according to Configuration 5, wherein a capacitance of the fifth capacitor is less than a capacitance of the first capacitor and less than a capacitance of the second capacitor.

Configuration 7

- The coupler according to any one of Configurations 1 to 5, wherein
- capacitances of the first and second capacitors are greater than 0.1 times capacitances of the third and fourth capacitors.

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Configuration 8

- The coupler according to any one of Configurations 1 to 7, further comprising:
- a first conductive member configured to apply a magnetic field to the space,
 - a coupling strength between the first nonlinear resonator and the second nonlinear resonator changes according to magnetic flux in the space.

Configuration 9

- The coupler according to any one of Configurations 1 to 8, wherein
- the coupler can resonate in a plurality of modes,
 - resonant frequencies of the plurality of modes are higher than a resonant frequency of the first nonlinear resonator, higher than a resonant frequency of the second nonlinear resonator, and lower than a sum of the resonant frequency of the first nonlinear resonator and the resonant frequency of the second nonlinear resonator.

Configuration 10

- A calculating device, comprising:
- the coupler according to any one of Configurations 1 to 7;
 - the first nonlinear resonator; and
 - the second nonlinear resonator.

Configuration 11

The calculating device according to Configuration 10, wherein

- the first nonlinear resonator includes:
 - a first nonlinear resonator Josephson junction; and
 - a first resonator capacitor,
- an end portion of the first nonlinear resonator Josephson junction and an end portion of the first resonator capacitor are electrically connected with the third capacitor end portion,
- an other end portion of the first nonlinear resonator Josephson junction and an other end portion of the first resonator capacitor are electrically connected with the first capacitor other-end portion,
- the second nonlinear resonator includes:
 - a second nonlinear resonator Josephson junction; and
 - a second resonator capacitor,
- an end portion of the second nonlinear resonator Josephson junction and an end portion of the second resonator capacitor are electrically connected with the fourth capacitor end portion, and
- an other end portion of the second nonlinear resonator Josephson junction and an other end portion of the second resonator capacitor are electrically connected with the second capacitor other-end portion.

Configuration 12

The calculating device according to Configuration 11, wherein

- a capacitance of the first capacitor is greater than 0.1 times a capacitance of the first resonator capacitor, and
- a capacitance of the second capacitor is greater than 0.1 times a capacitance of the second resonator capacitor.

Configuration 13

The calculating device according to Configuration 12, wherein

- a capacitance of the third capacitor is greater than 0.1 times the capacitance of the first resonator capacitor, and
- a capacitance of the fourth capacitor is greater than 0.1 times the capacitance of the second resonator capacitor.

Configuration 14

The calculating device according to any one of Configurations 10 to 13, further comprising:

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a controller,
the coupler further including a first magnetic field application part configured to apply a magnetic field to the space,
the controller being configured to change a magnetic flux in the space by controlling the first magnetic field application part.

Configuration 15

The calculating device according to Configuration 14, wherein

the first magnetic field application part includes a first conductive member, and
the controller changes the magnetic flux by modulating a current supplied to the first conductive member.

Configuration 16

The calculating device according to Configuration 14 or Configuration 15, wherein

a two-qubit gate operation for the first and second nonlinear resonators is performed by the change of the magnetic flux.

Configuration 17

The calculating device according to Configuration 14 or Configuration 15, wherein

the controller is configured to change a coupling strength between the first nonlinear resonator and the second nonlinear resonator by controlling the magnetic flux in the space.

Configuration 18

The calculating device according to Configuration 14 or Configuration 15, wherein

the controller can substantially decouple the first nonlinear resonator from the second nonlinear resonator by controlling the magnetic flux in the space.

Configuration 19

The calculating device according to any one of Configurations 10 to 18, wherein

the coupler can resonate in a plurality of modes, and resonant frequencies of the plurality of modes are not more than 2 times a resonant frequency of the first nonlinear resonator and not more than 2 times a resonant frequency of the second nonlinear resonator.

Configuration 20

The calculating device according to any one of Configurations 10 to 18, wherein

the coupler can resonate in a plurality of modes, and resonant frequencies of the plurality of modes are higher than a resonant frequency of the first nonlinear resonator, higher than a resonant frequency of the second nonlinear resonator, and lower than a sum of the resonant frequency of the first nonlinear resonator and the resonant frequency of the second nonlinear resonator.

According to embodiments, a coupler and a calculating device can be provided in which the controllability can be improved.

Hereinabove, exemplary embodiments of the invention are described with reference to specific examples. However, the embodiments of the invention are not limited to these specific examples. For example, one skilled in the art may similarly practice the invention by appropriately selecting specific configurations of components included in couplers and calculating devices such as nonlinear resonators, inductors, capacitors, conductive members, etc., from known art. Such practice is included in the scope of the invention to the extent that similar effects thereto are obtained.

Further, any two or more components of the specific examples may be combined within the extent of technical

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feasibility and are included in the scope of the invention to the extent that the purport of the invention is included.

Moreover, all couplers, and calculating devices practicable by an appropriate design modification by one skilled in the art based on the couplers, and the calculating devices described above as embodiments of the invention also are within the scope of the invention to the extent that the purport of the invention is included.

Various other variations and modifications can be conceived by those skilled in the art within the spirit of the invention, and it is understood that such variations and modifications are also encompassed within the scope of the invention.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention.

What is claimed is:

1. A coupler, comprising:

a first capacitor including a first capacitor end portion and a first capacitor other-end portion;

a first inductor including

a first inductor end portion electrically connected with the first capacitor end portion, and

a first inductor other-end portion;

a second capacitor including a second capacitor end portion and a second capacitor other-end portion;

a second inductor including

a second inductor end portion electrically connected with the second capacitor end portion, and

a second inductor other-end portion electrically connected with the first capacitor other-end portion, the first inductor other-end portion, and the second capacitor other-end portion; and

a first Josephson junction including

a first Josephson junction end portion electrically connected with the first capacitor end portion, and

a first Josephson junction other-end portion electrically connected with the second capacitor end portion, the first capacitor end portion being couplable with a first nonlinear resonator, and the second capacitor end portion being couplable with a second nonlinear resonator.

2. The coupler according to claim 1, wherein

the first inductor includes a first conductive layer having a meandering structure, and

the second inductor includes a second conductive layer having a meandering structure.

3. The coupler according to claim 1, wherein

the first inductor includes a first inductor Josephson junction, and

the second inductor includes a second inductor Josephson junction.

4. The coupler according to claim 1, wherein

the first inductor includes a plurality of first inductor Josephson junctions,

the plurality of first inductor Josephson junctions is electrically connected in series to each other,

the second inductor includes a plurality of second inductor Josephson junctions, and

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the plurality of second inductor Josephson junctions is electrically connected in series to each other.

5. The coupler according to claim 1, further comprising: a fifth capacitor including a fifth capacitor end portion and a fifth capacitor other-end portion, the fifth capacitor end portion being electrically connected with the first capacitor end portion, the fifth capacitor other-end portion being electrically connected with the second capacitor end portion.

6. The coupler according to claim 5, wherein a capacitance of the fifth capacitor is less than a capacitance of the first capacitor and less than a capacitance of the second capacitor.

7. The coupler according to claim 1, further comprising: a first conductive member configured to apply a magnetic field to a space being surrounded with the first inductor, the second inductor, and the first Josephson junction, and a coupling strength between the first nonlinear resonator and the second nonlinear resonator changes according to magnetic flux in the space.

8. The coupler according to claim 1, wherein the coupler is capable of resonating in a plurality of modes, at least two of resonant frequencies in the modes are higher than a resonant frequency of the first nonlinear resonator, are higher than a resonant frequency of the second nonlinear resonator, and are lower than a sum of the resonant frequency of the first nonlinear resonator and the resonant frequency of the second nonlinear resonator, the first nonlinear resonator and the second nonlinear resonator can be substantially decoupled by controlling magnetic flux in a space being surrounded with the first inductor, the second inductor, and the first Josephson junction, and in a state of being substantially decoupled, the resonant frequencies in the modes are higher than the resonant frequency of the first nonlinear resonator, and are higher than the resonant frequency of the second nonlinear resonator.

9. The coupler according to claim 1, wherein the first nonlinear resonator consists of a first nonlinear resonator Josephson junction and a first resonator capacitor, and the second nonlinear resonator consists of a second nonlinear resonator Josephson junction and a second resonator capacitor.

10. A calculating device, comprising: the coupler according to claim 1; the first nonlinear resonator; and the second nonlinear resonator.

11. The device according to claim 10, wherein the first nonlinear resonator includes: a first nonlinear resonator Josephson junction; and a first resonator capacitor, an end portion of the first nonlinear resonator Josephson junction and an end portion of the first resonator capacitor are electrically connected with the first capacitor end portion,

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an other end portion of the first nonlinear resonator Josephson junction and an other end portion of the first resonator capacitor are electrically connected with the first capacitor other-end portion,

the second nonlinear resonator includes: a second nonlinear resonator Josephson junction; and a second resonator capacitor, an end portion of the second nonlinear resonator Josephson junction and an end portion of the second resonator capacitor are electrically connected with the second capacitor end portion, and an other end portion of the second nonlinear resonator Josephson junction and an other end portion of the second resonator capacitor are electrically connected with the second capacitor other-end portion.

12. The device according to claim 11, wherein a capacitance of the first capacitor is greater than 0.1 times a capacitance of the first resonator capacitor, and a capacitance of the second capacitor is greater than 0.1 times a capacitance of the second resonator capacitor.

13. The device according to claim 10, further comprising: a controller, the coupler further including a first magnetic field application part configured to apply a magnetic field to a space being surrounded with the first inductor, the second inductor, and the first Josephson junction, the controller being configured to change a magnetic flux in the space by controlling the first magnetic field application part.

14. The device according to claim 13, wherein the first magnetic field application part includes a first conductive member, and the controller is configured to change the magnetic flux by modulating a current supplied to the first conductive member.

15. The device according to claim 13, wherein a two-qubit gate operation for the first and second nonlinear resonators is performed by the change of the magnetic flux.

16. The device according to claim 13, wherein the controller is configured to change a coupling strength between the first nonlinear resonator and the second nonlinear resonator by controlling the magnetic flux in the space.

17. The device according to claim 13, wherein the controller is configured to substantially decouple the first nonlinear resonator from the second nonlinear resonator by controlling the magnetic flux in the space.

18. The device according to claim 10, wherein the coupler is capable of resonating in a plurality of modes, and resonant frequencies of the plurality of modes are not more than 2 times a resonant frequency of the first nonlinear resonator and not more than 2 times a resonant frequency of the second nonlinear resonator.

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